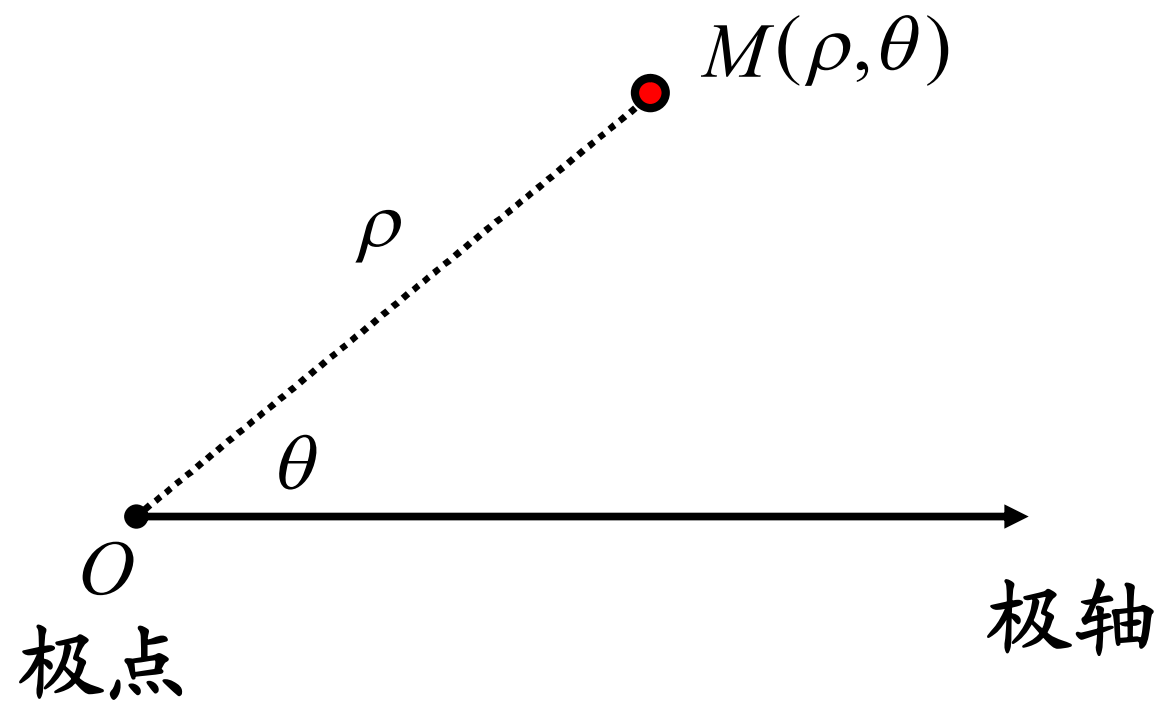


极坐标曲线



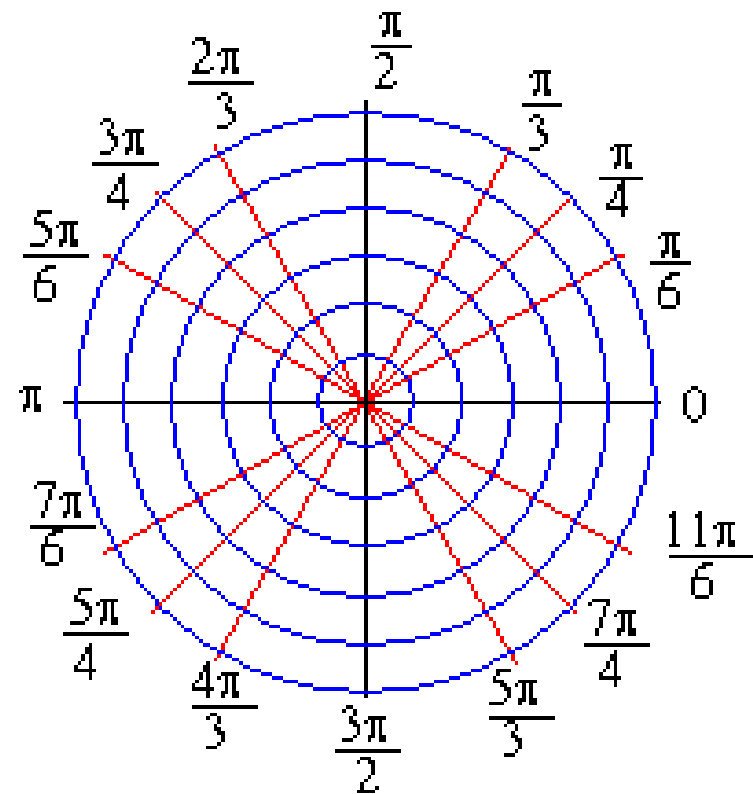
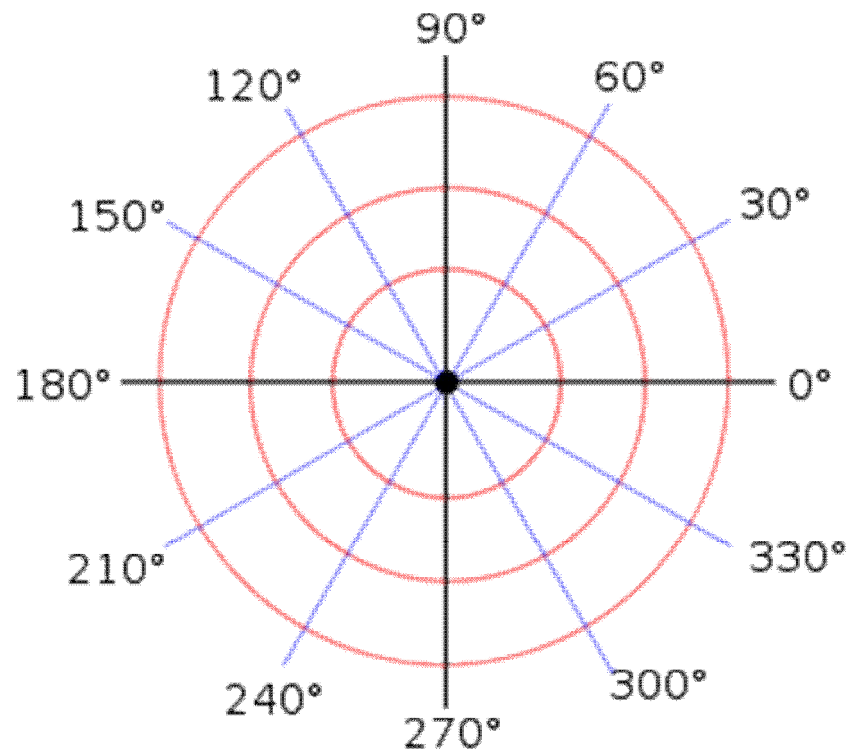
极坐标

点的极坐标

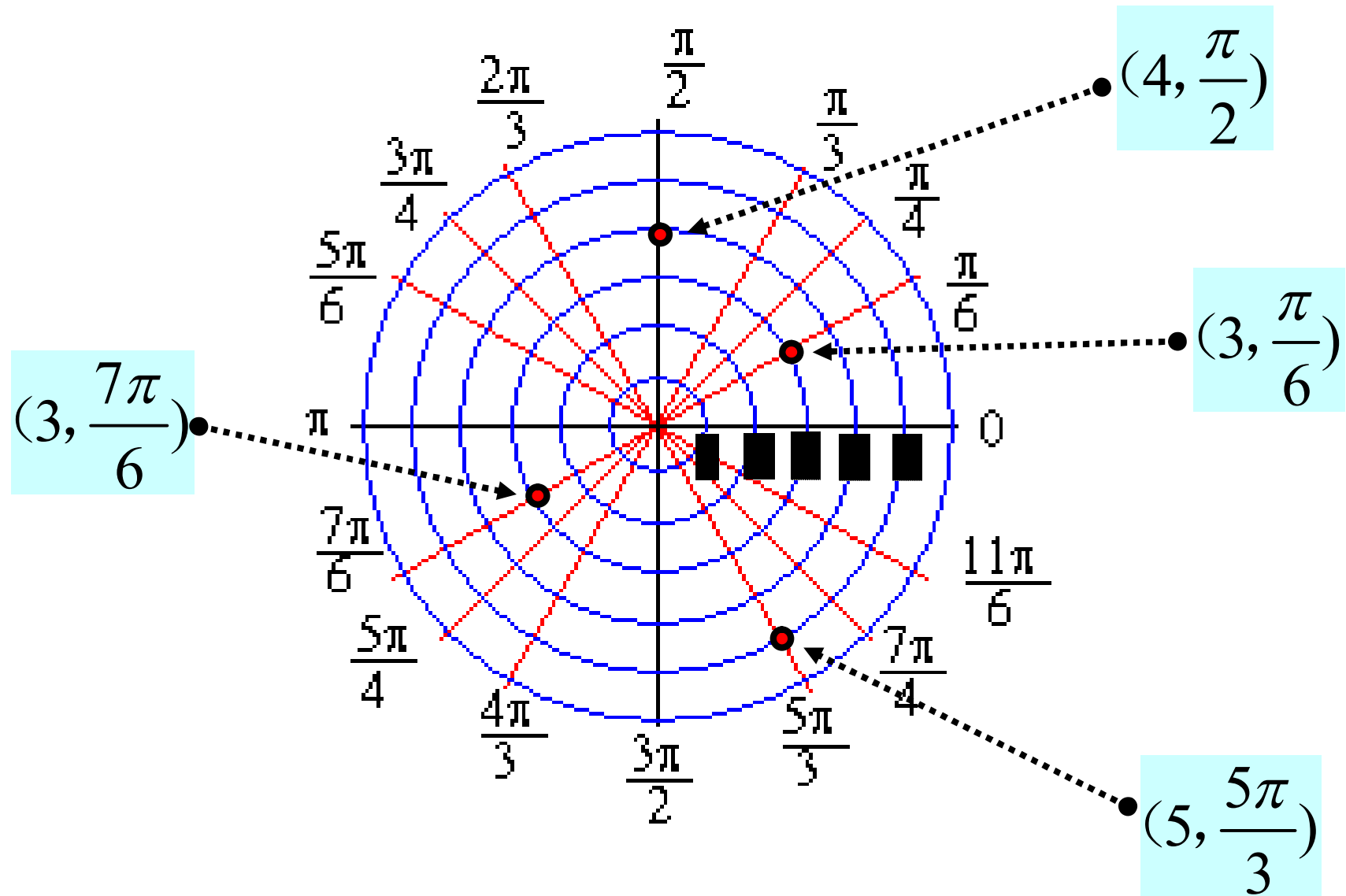


极坐标系

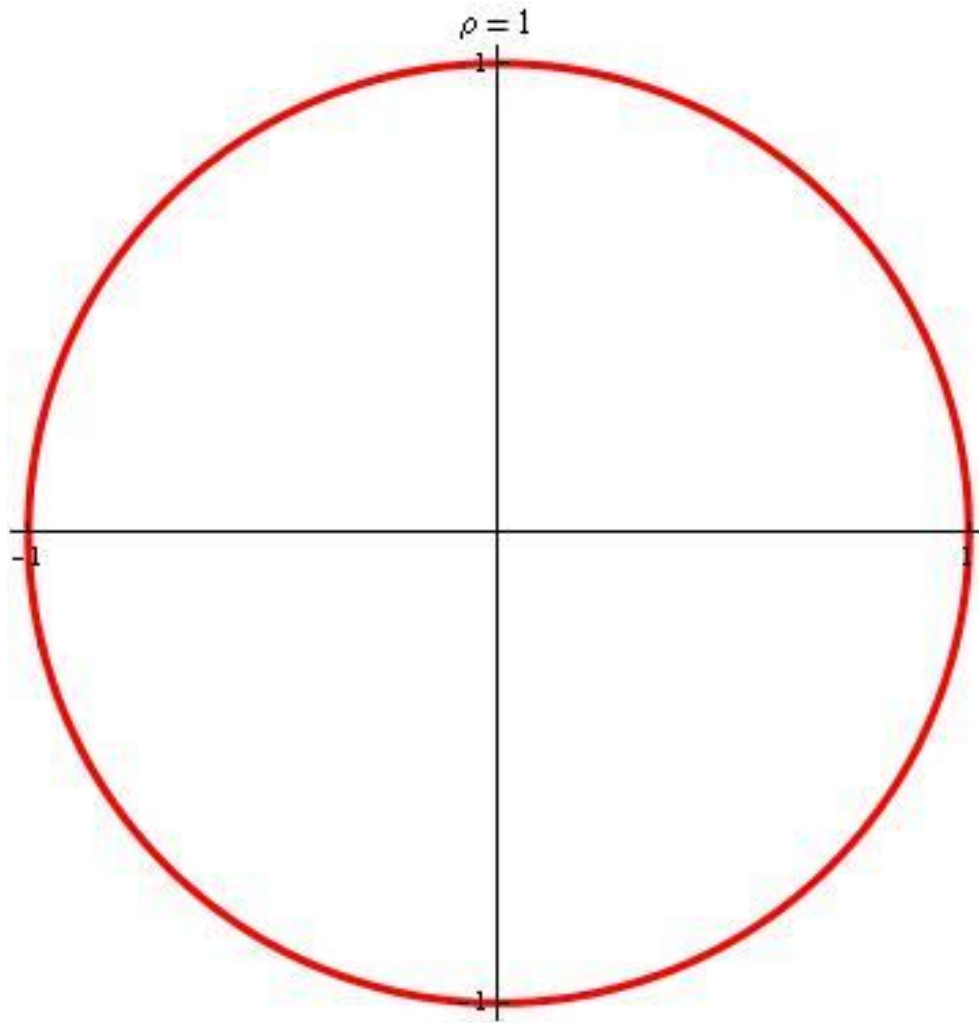
polar coordinate system



点的极坐标

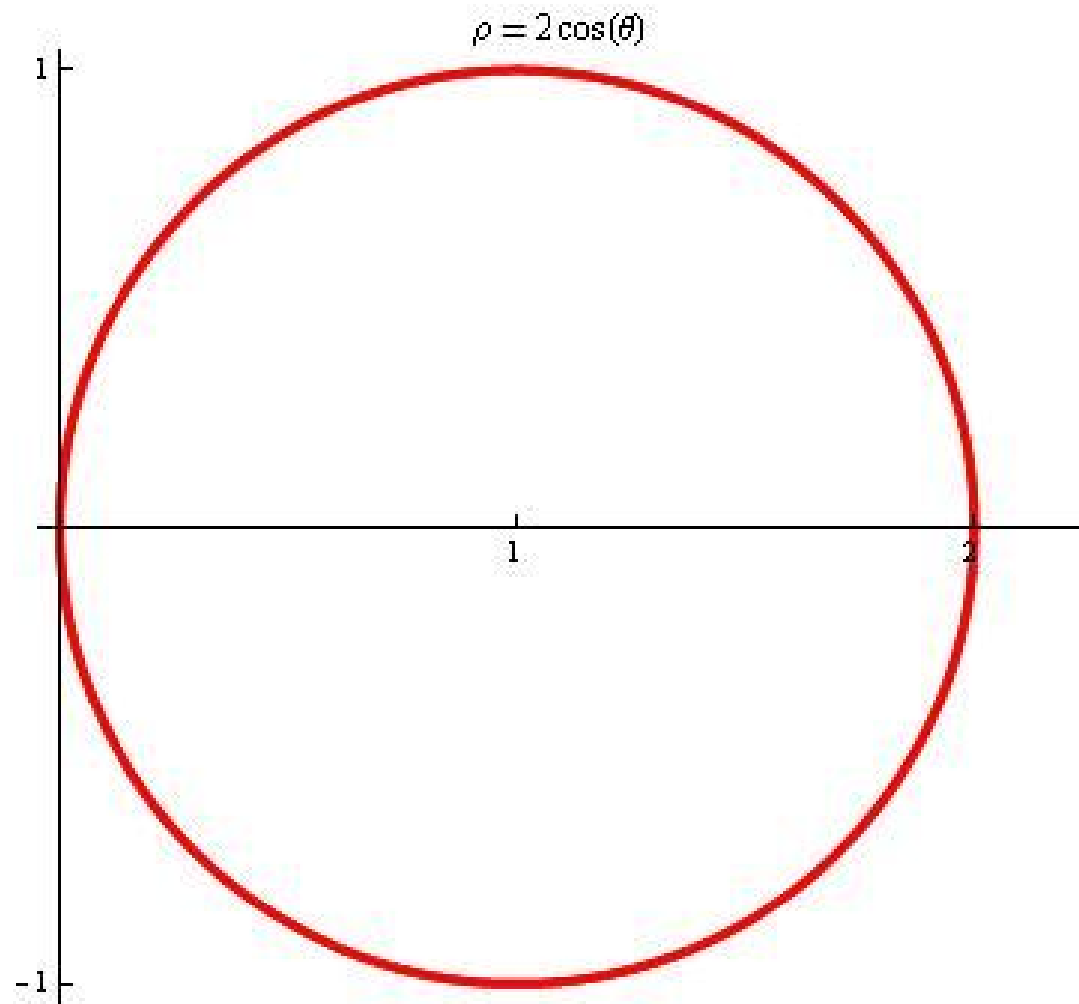


极坐标曲线



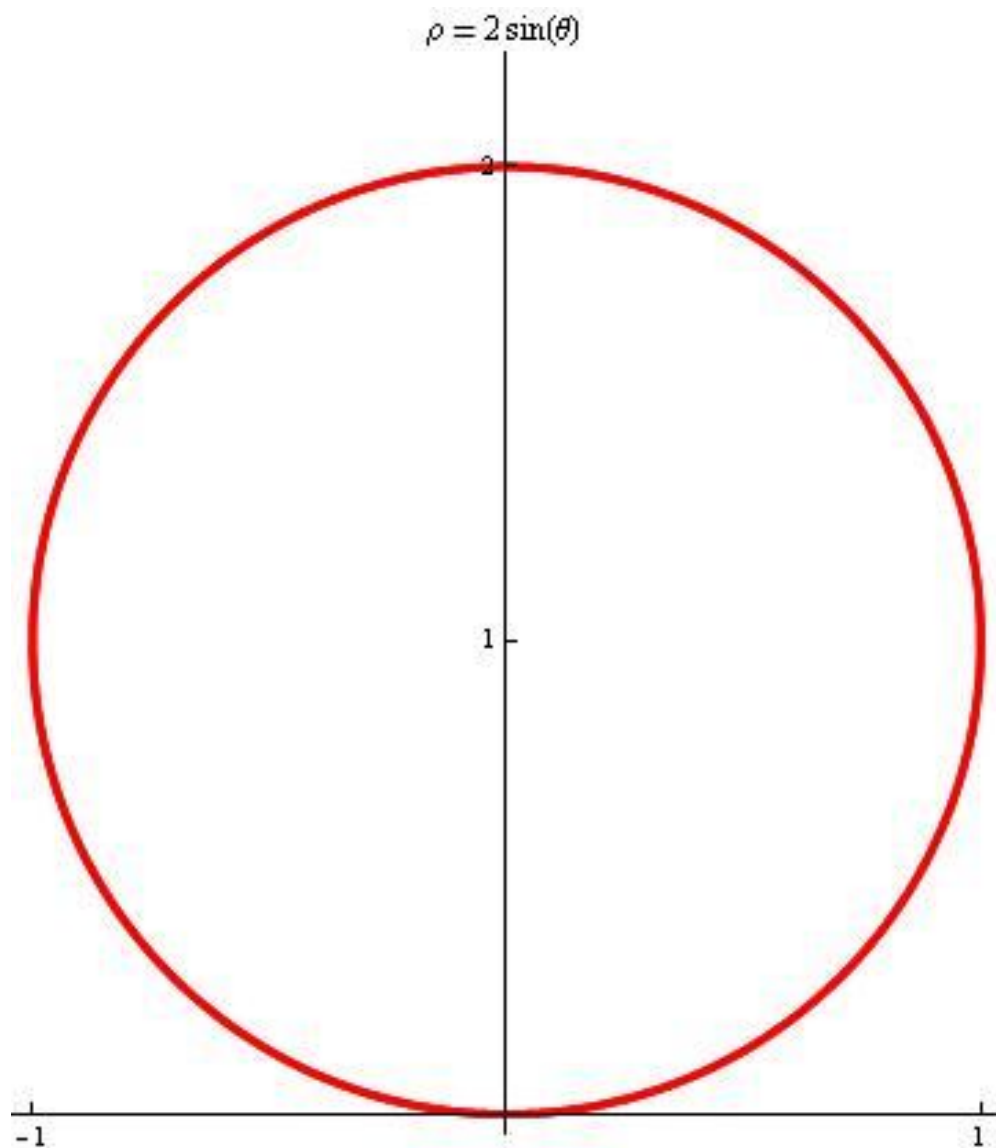
圆 (**Circle**)

$$\rho = a$$



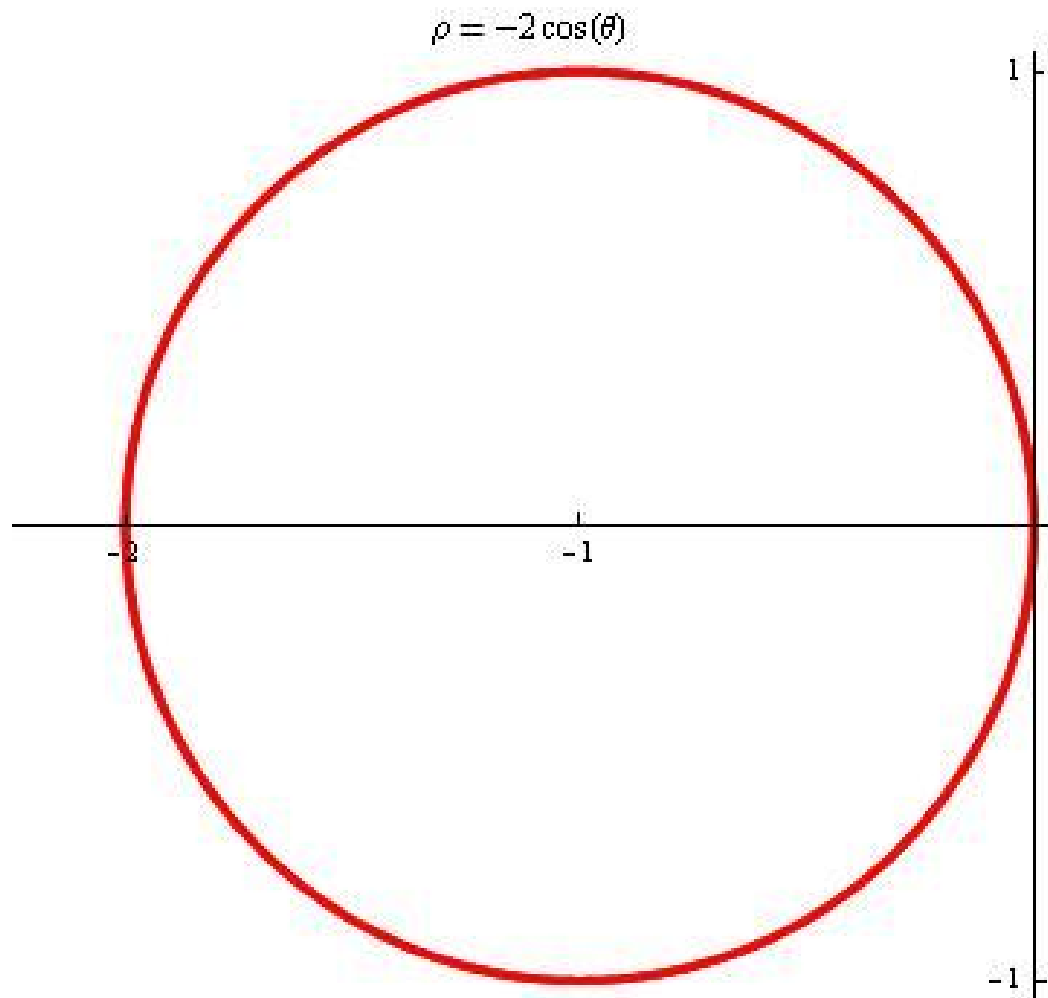
圆 (Circle)

$$\rho = 2a \cos \theta$$



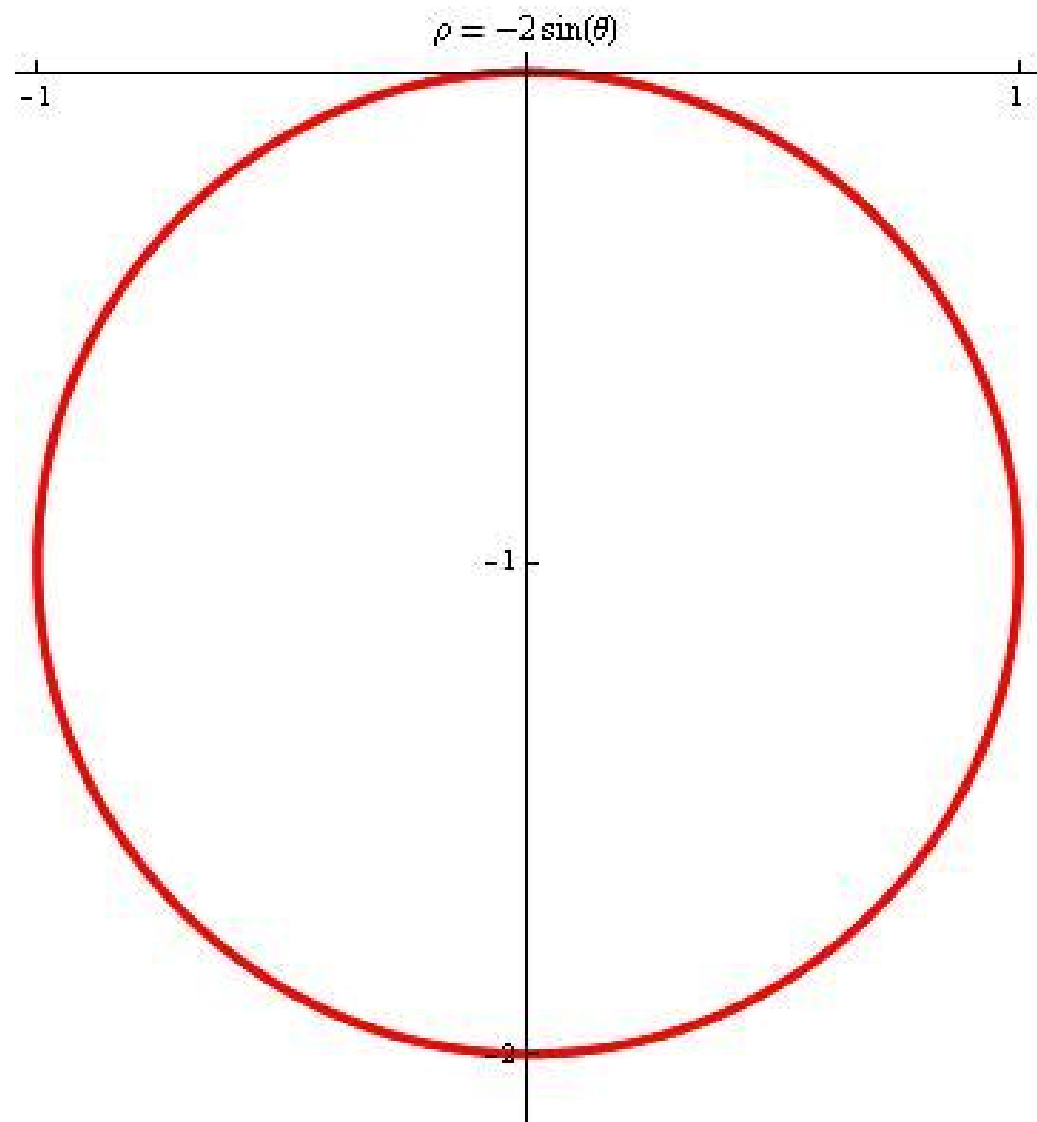
圆 (Circle)

$$\rho = 2a \sin \theta$$



圆 (Circle)

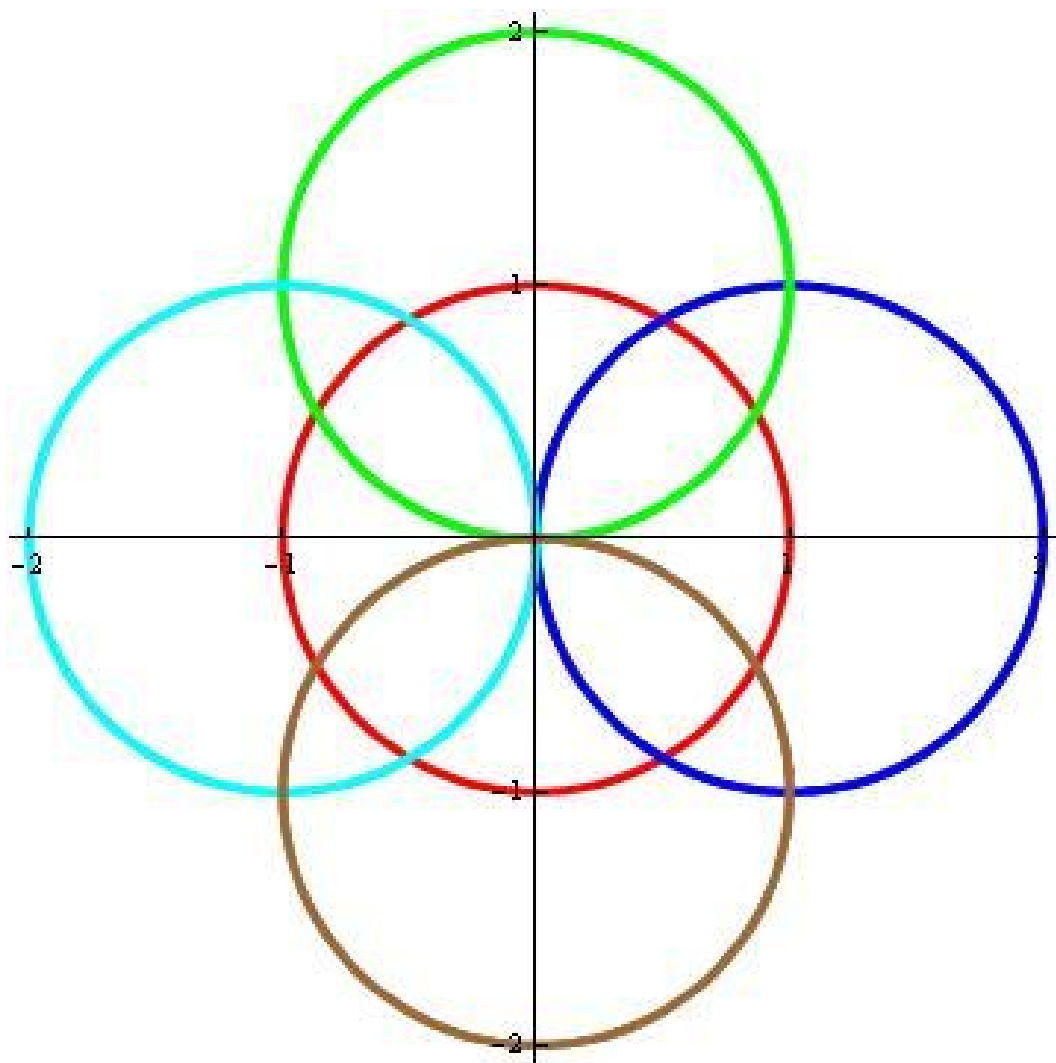
$$\rho = -2a \cos \theta$$



圆 (Circle)

$$\rho = -2a \sin \theta$$

五个圆



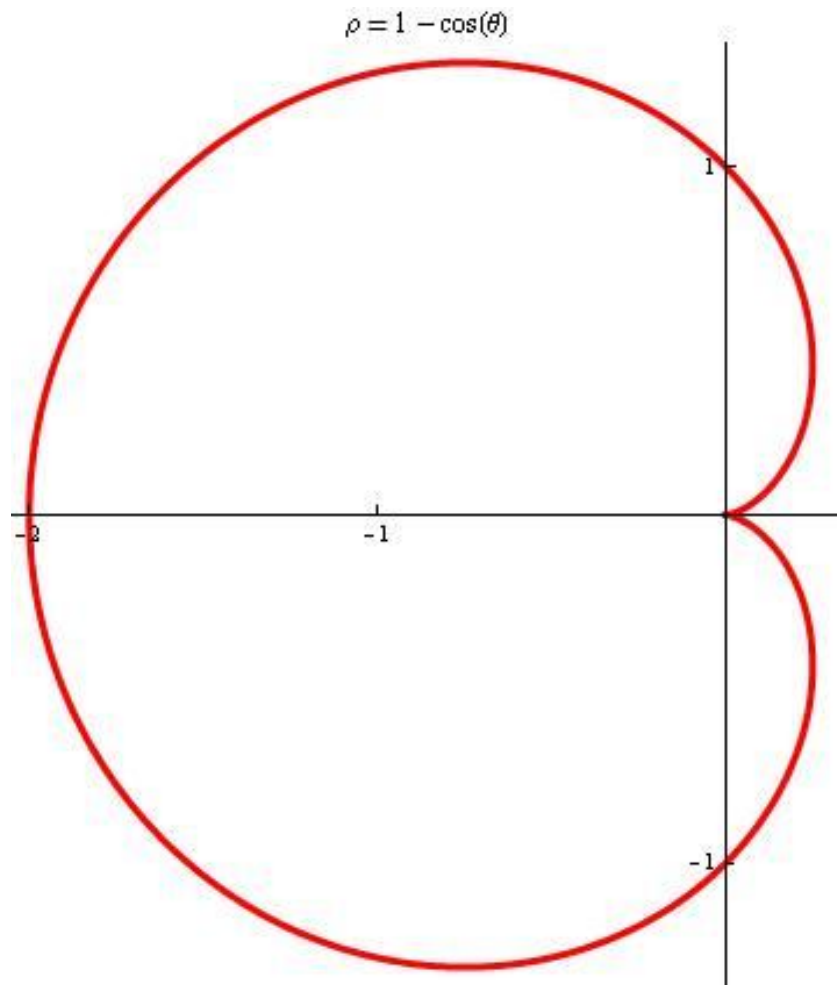
$$\rho = a$$

$$\rho = 2a \cos \theta$$

$$\rho = 2a \sin \theta$$

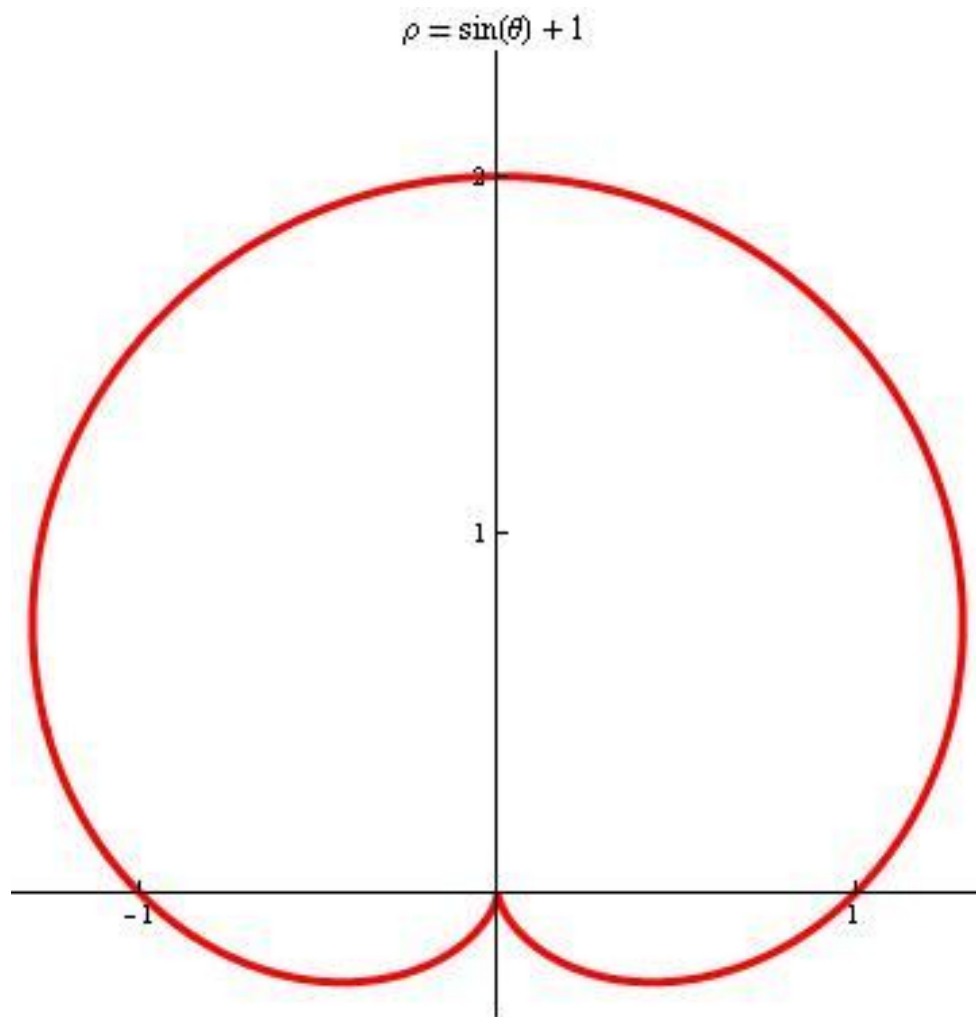
$$\rho = -2a \cos \theta$$

$$\rho = -2a \sin \theta$$



心形线
(**cardioid**)

$$\rho = a(1 - \cos \theta)$$



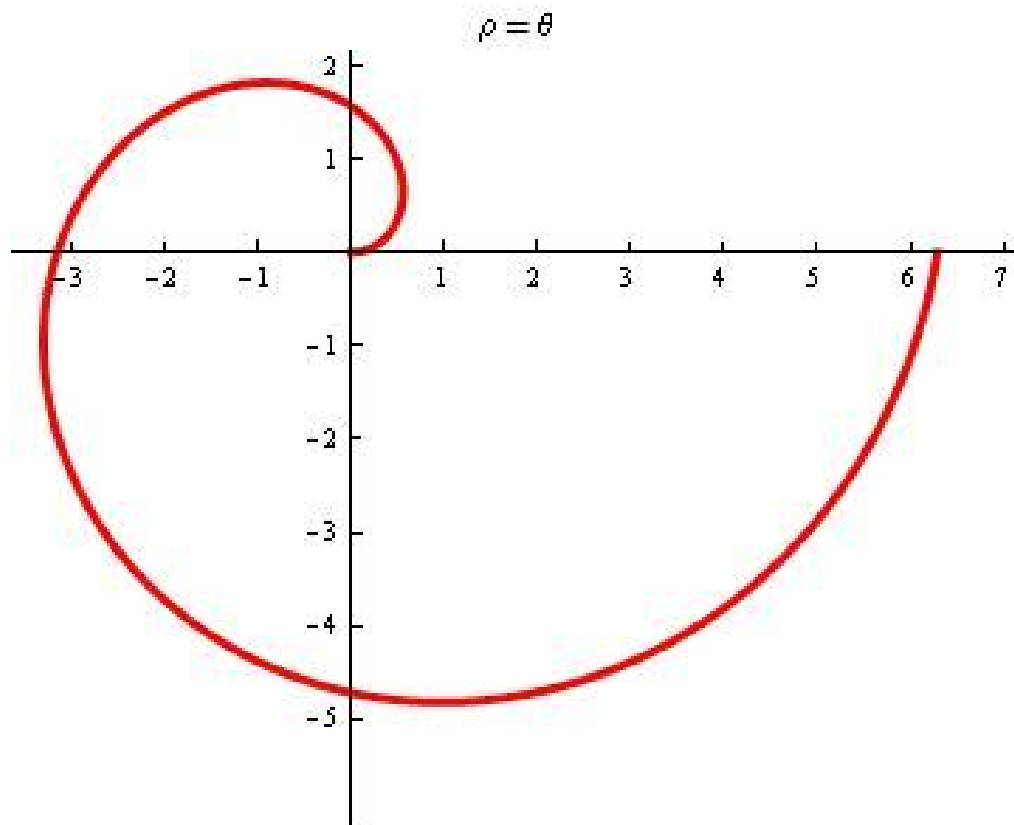
心形线
(**cardioid**)

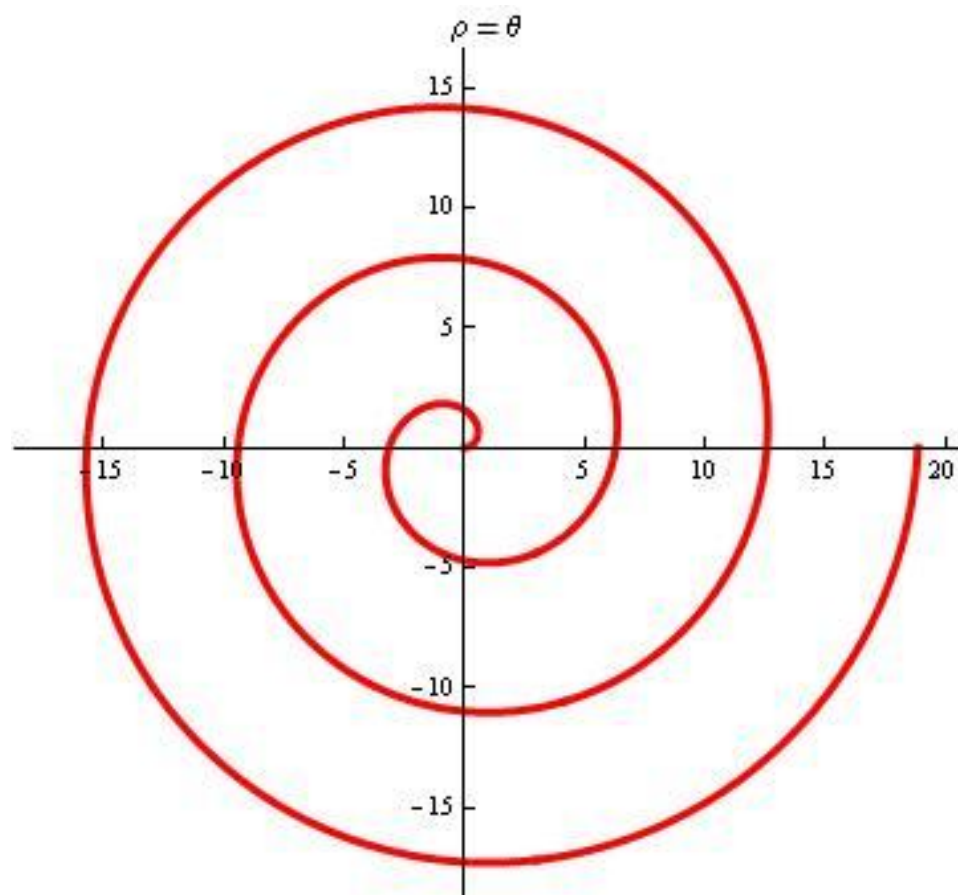
$$\rho = a(1 + \sin \theta)$$

阿基米德螺线 (Archimedes' spiral)

等速螺线

$$\rho = a\theta$$



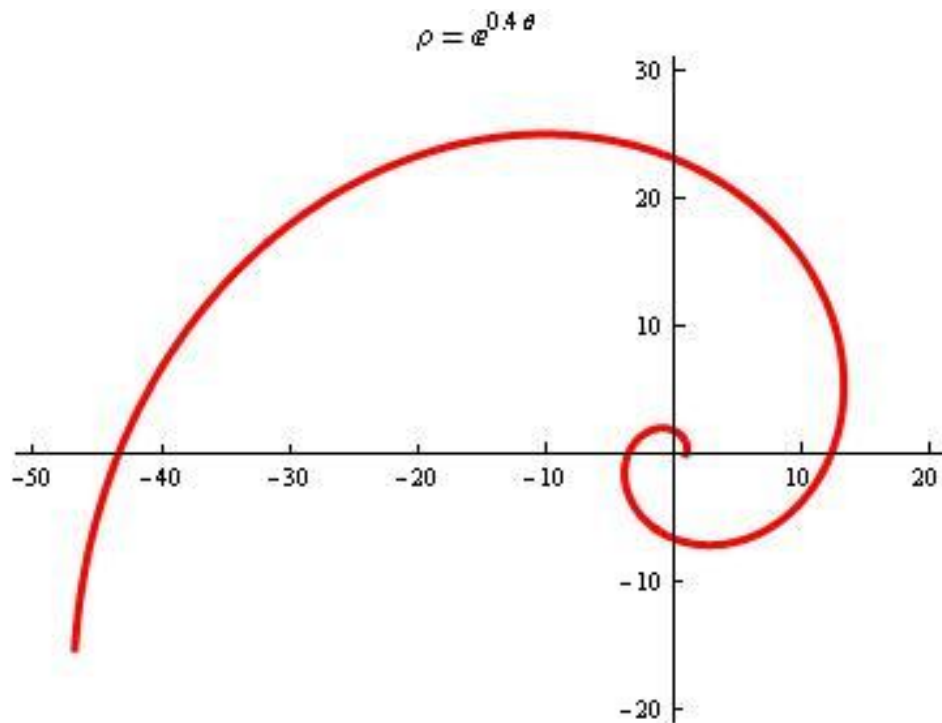


阿基米德螺线
(Archimedes'
spiral)
等速螺线

$$\rho = a\theta$$

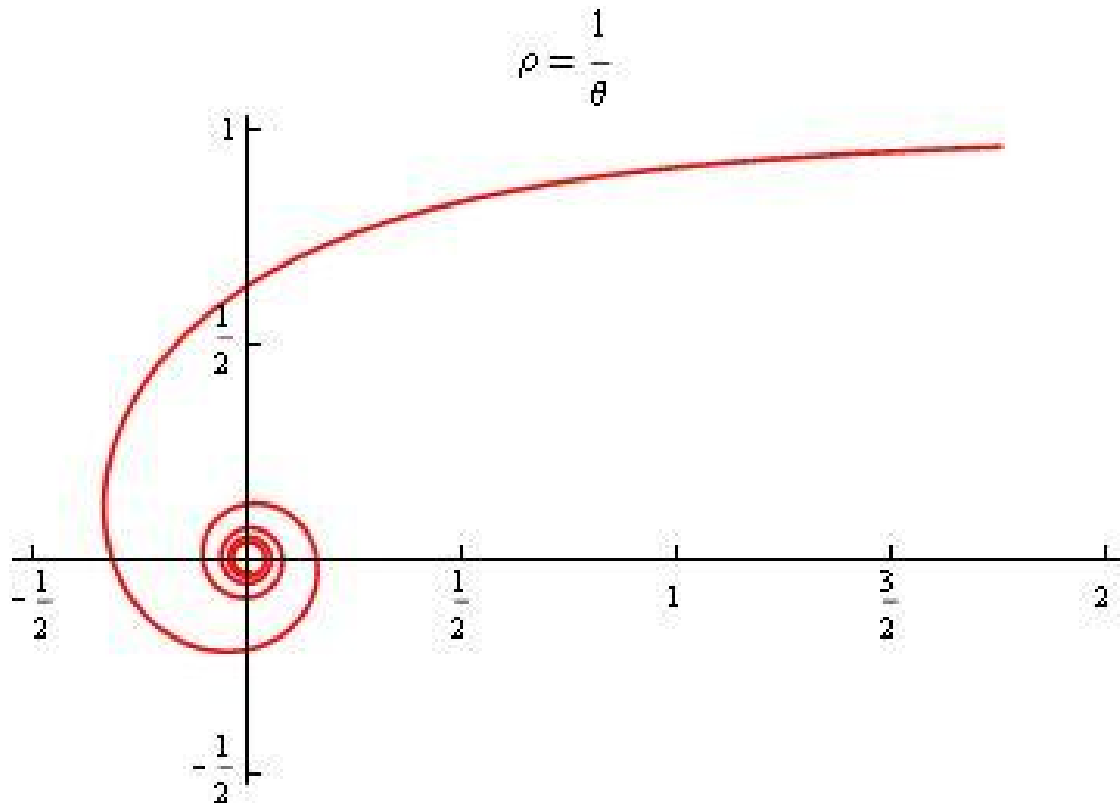
对数螺线 (Logarithmic spiral)

等角螺线



$$\rho = e^{a\theta}$$

双曲螺线(倒数螺线) (Hyperbolic spiral)

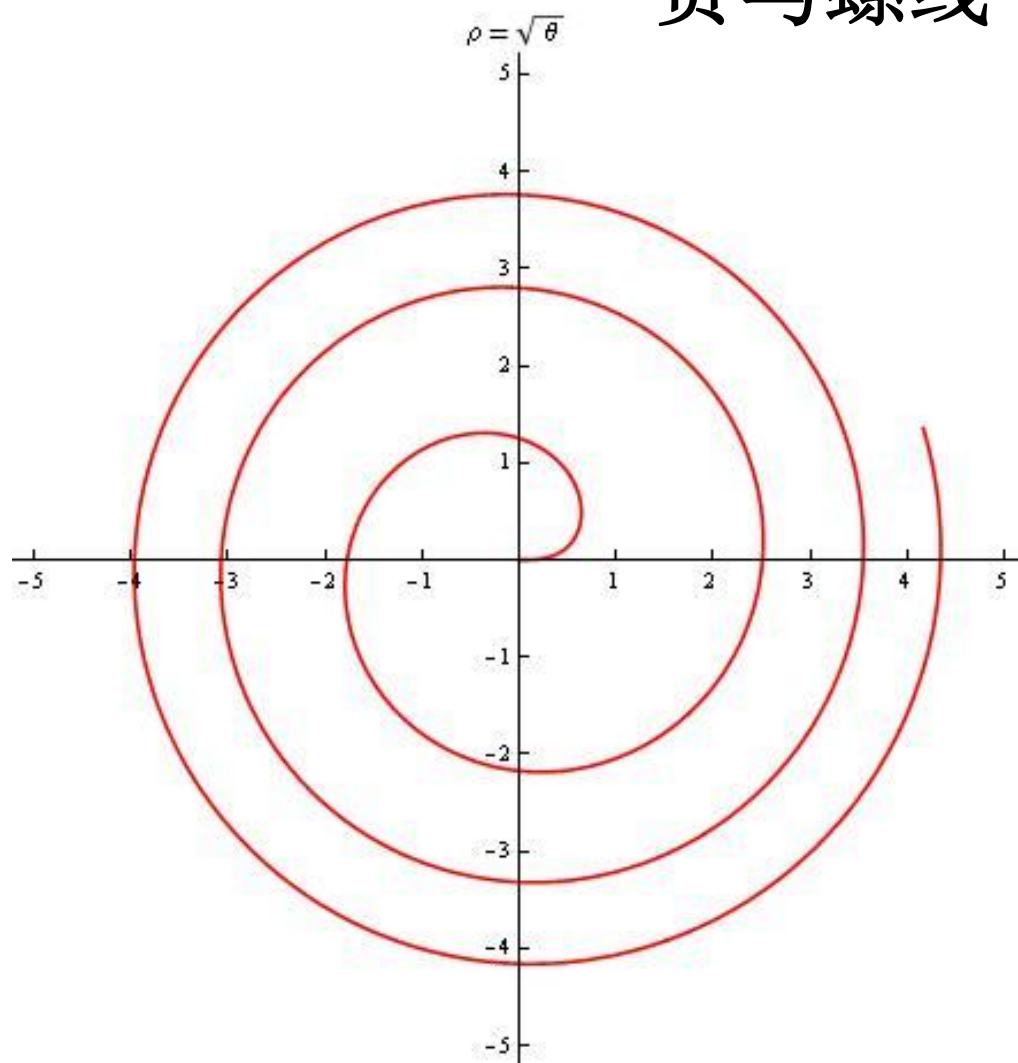


$$\rho = \frac{a}{\theta}$$

$$0.5 < \theta < 30$$

抛物螺线 (Parabolic spiral)

费马螺线 (Fermat's spiral)



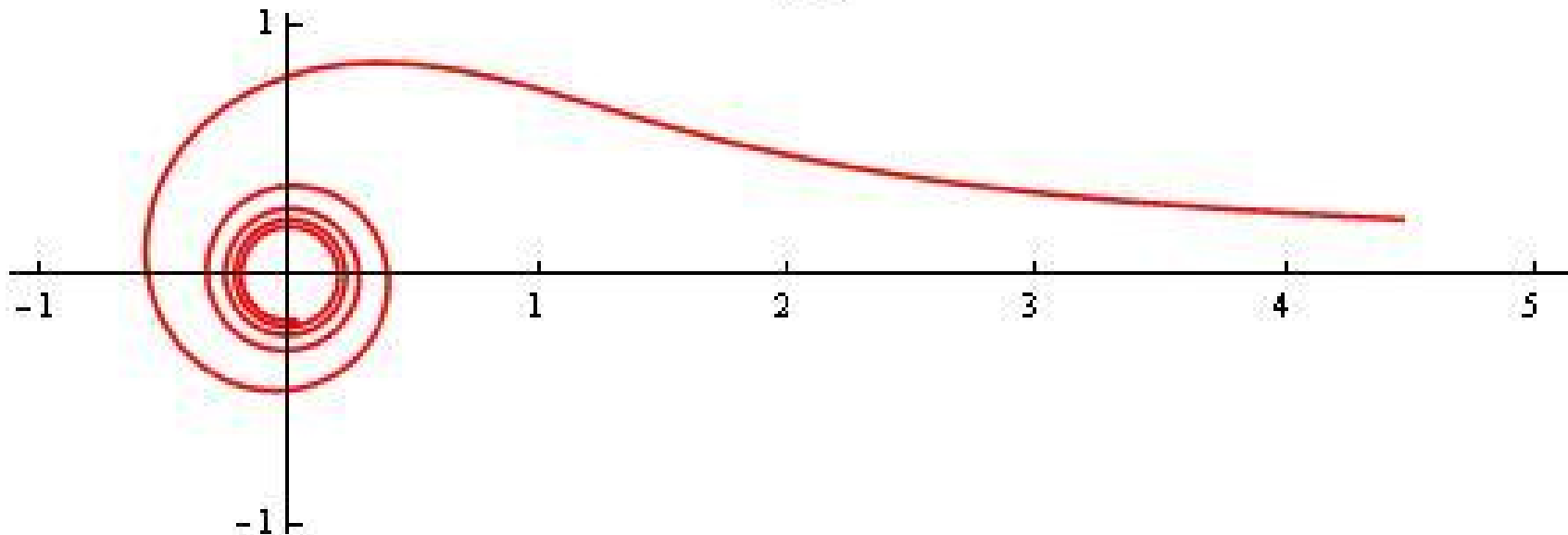
$$\rho^2 = a\theta$$

平方倒数螺线（连锁螺线）（Lituus）

$$\rho = \frac{a}{\sqrt{\theta}}$$

$$0.1 < \theta < 40$$

$$\rho = \frac{1}{\sqrt{\theta}}$$

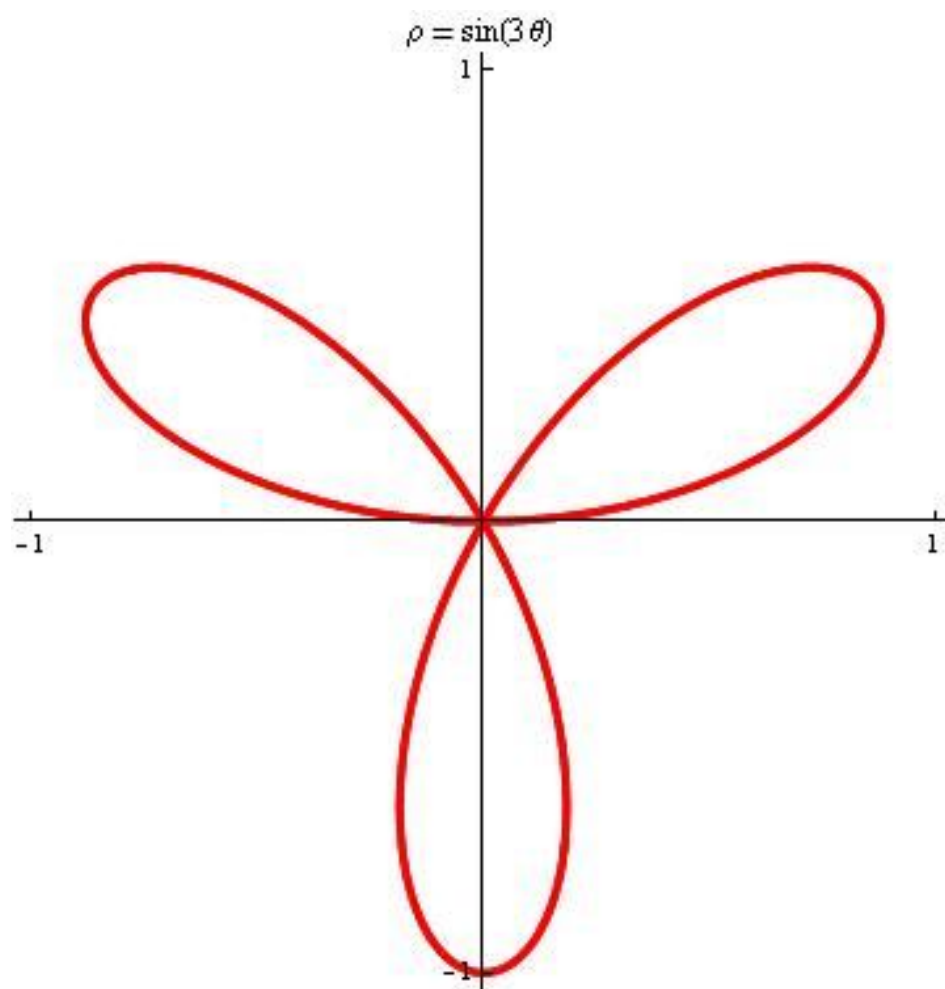


玫瑰线

三叶玫瑰线

**Three-leaved rose
curve**

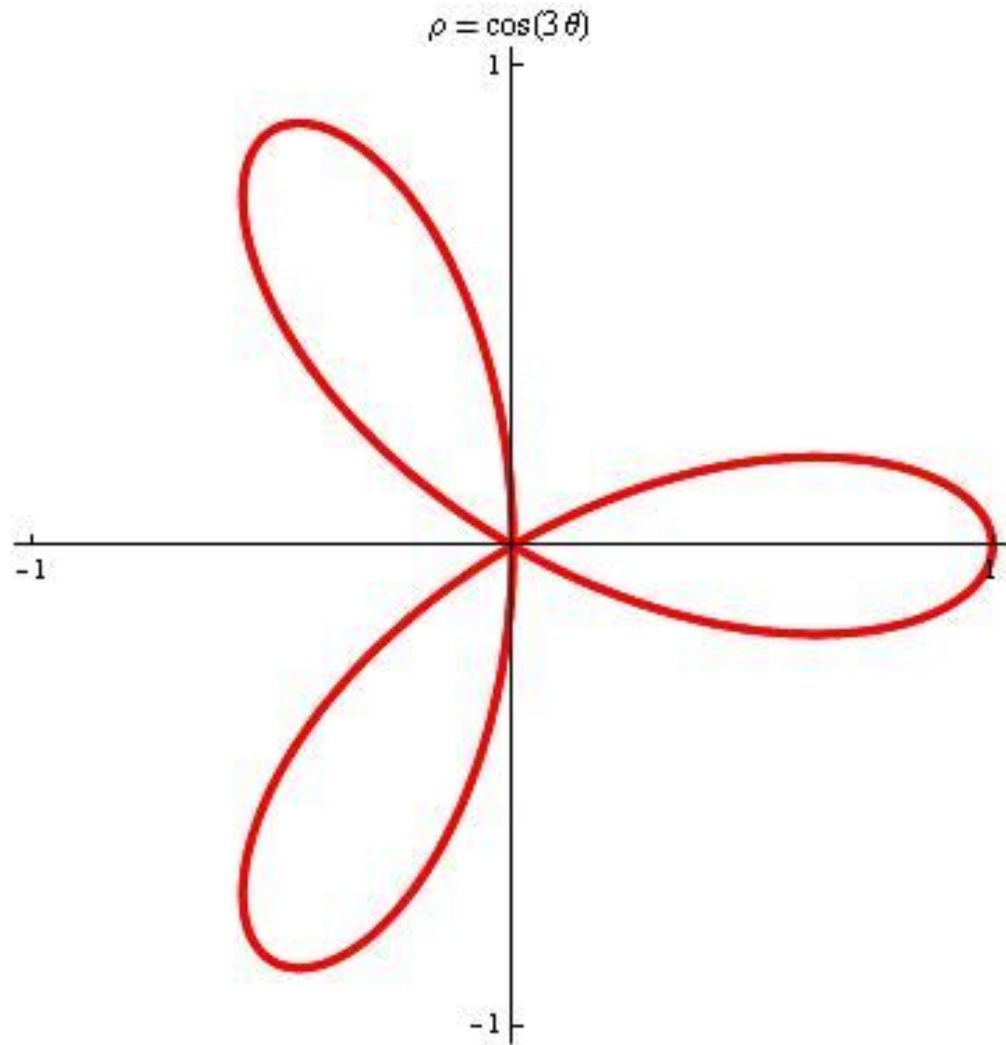
$$\rho = a \sin 3\theta$$



三叶玫瑰线

**Three-leaved rose
curve**

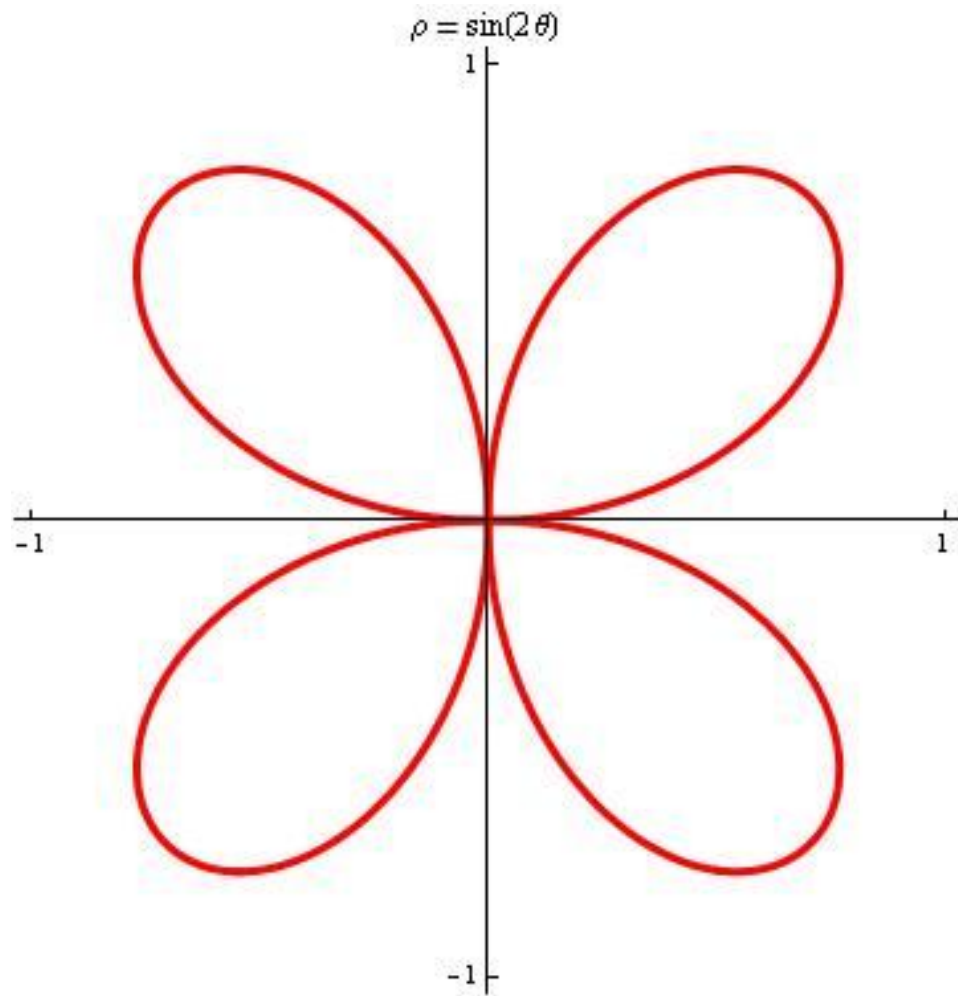
$$\rho = a \cos 3\theta$$



四叶玫瑰线

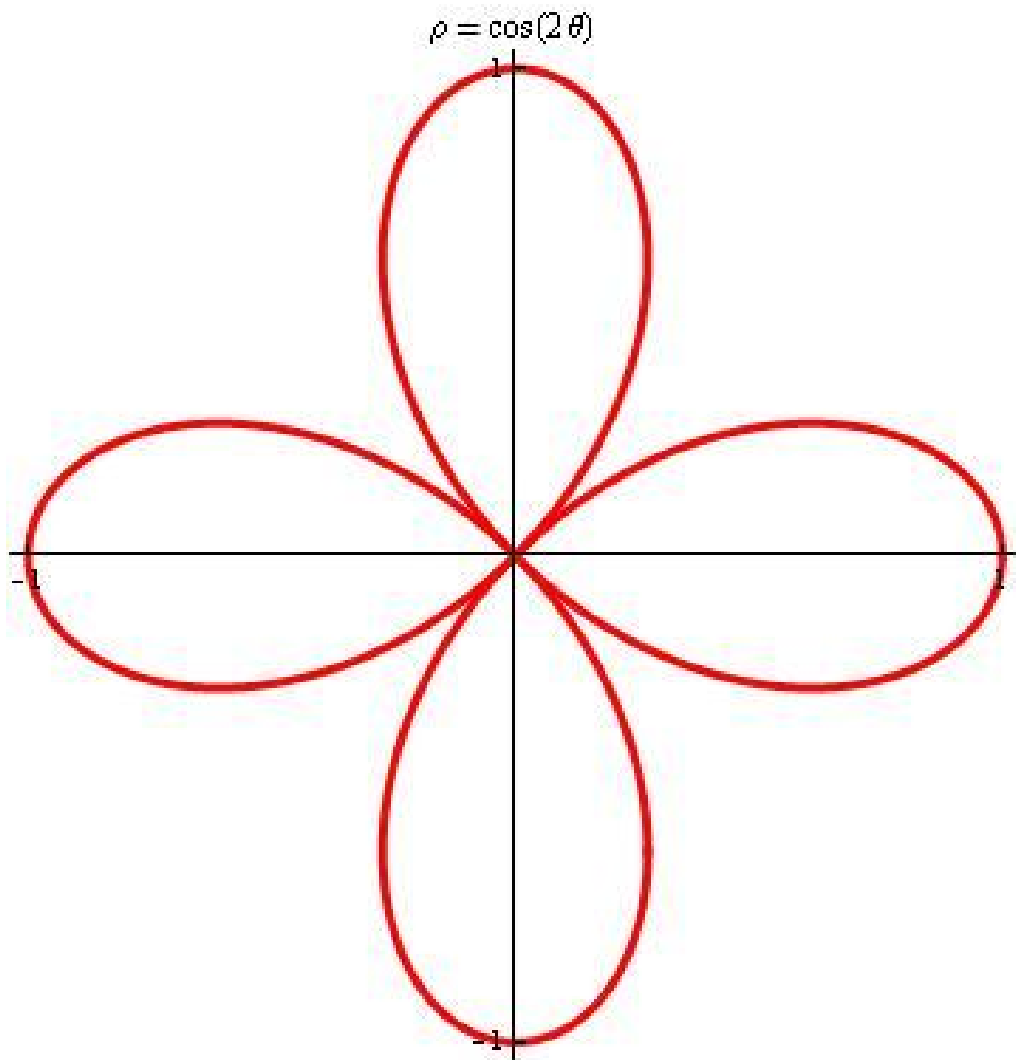
Four-leaved rose curve

$$\rho = a \sin 2\theta$$



四叶玫瑰线

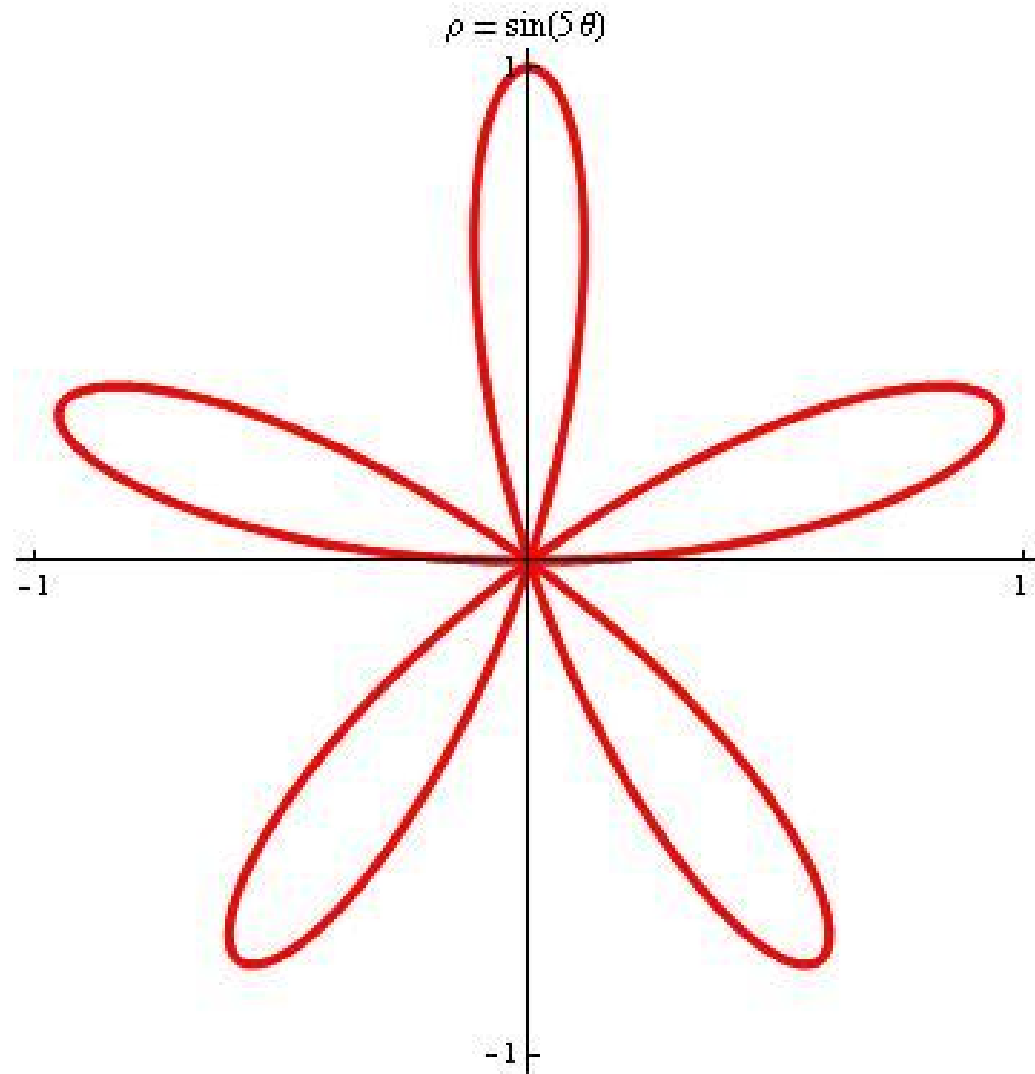
**Four-leaved rose
curve**



$$\rho = a \cos 2\theta$$

五叶玫瑰线

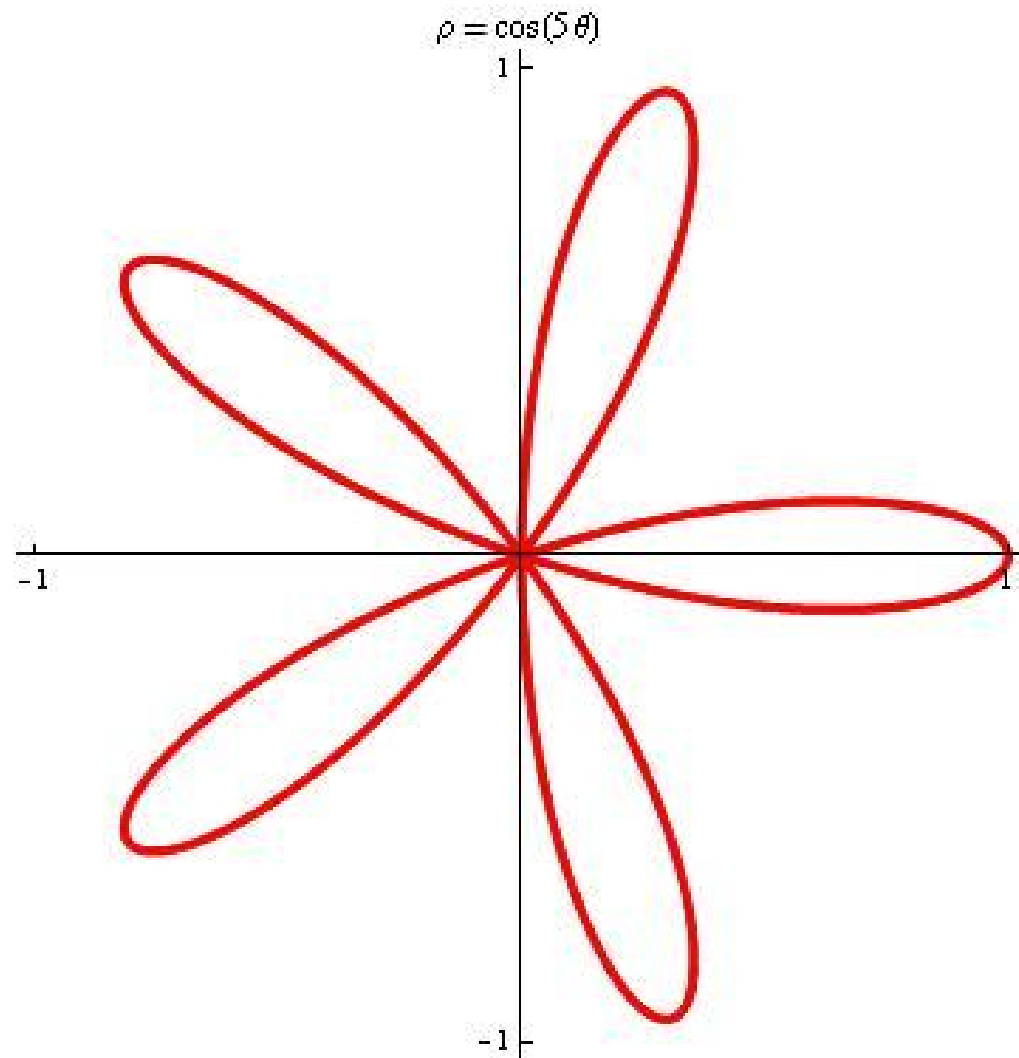
**Five-leaved rose
curve**



$$\rho = a \sin 5\theta$$

五叶玫瑰线

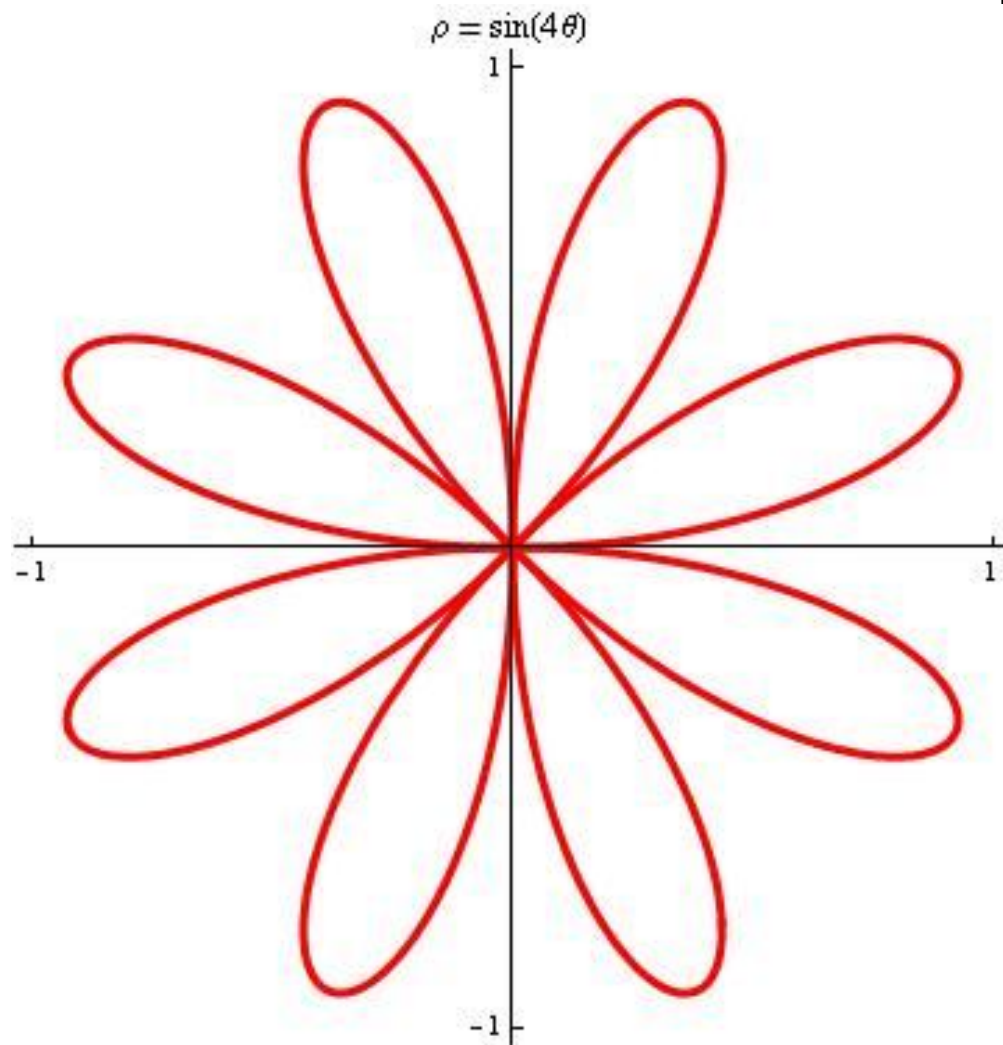
Five-leaved rose curve



$$\rho = a \cos 5\theta$$

八叶玫瑰线

Eight-leaved rose curve

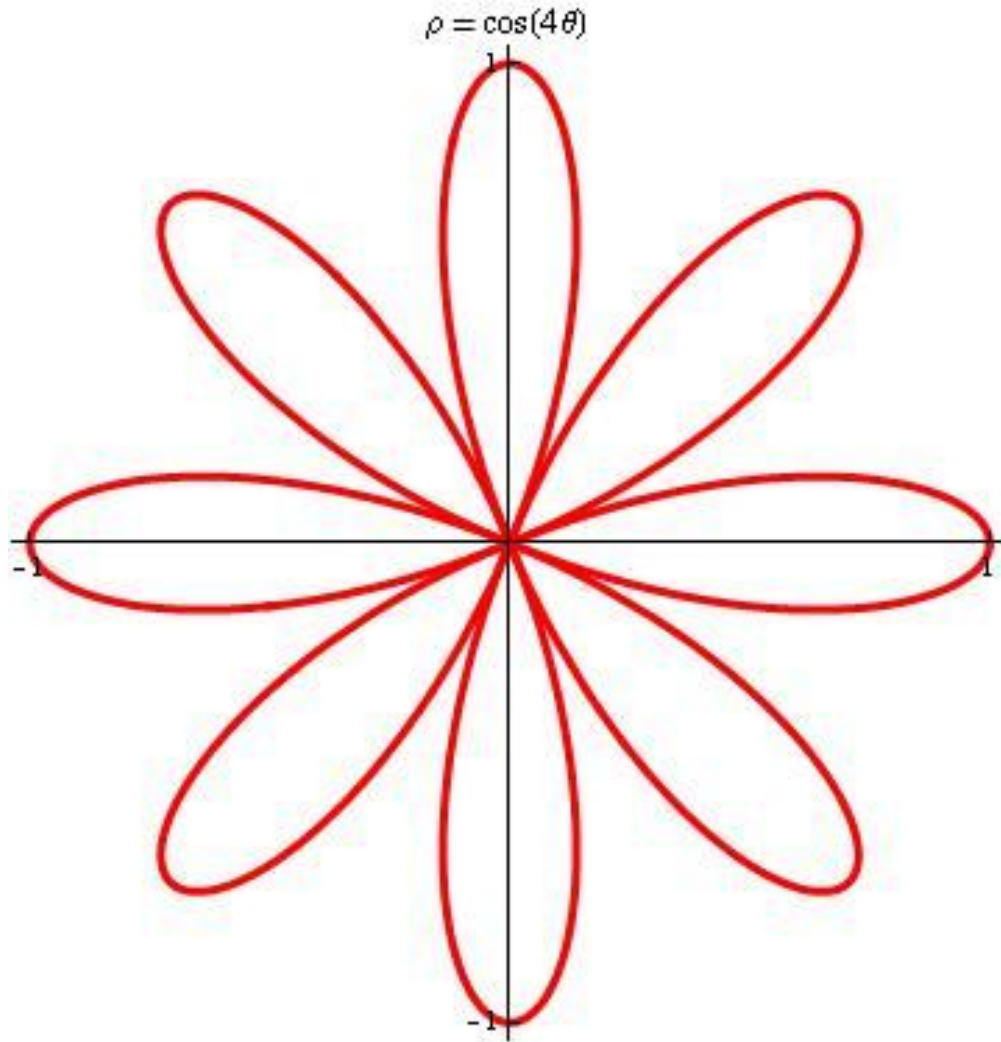


$$\rho = a \sin 4\theta$$

八叶玫瑰线

Eight-leaved rose curve

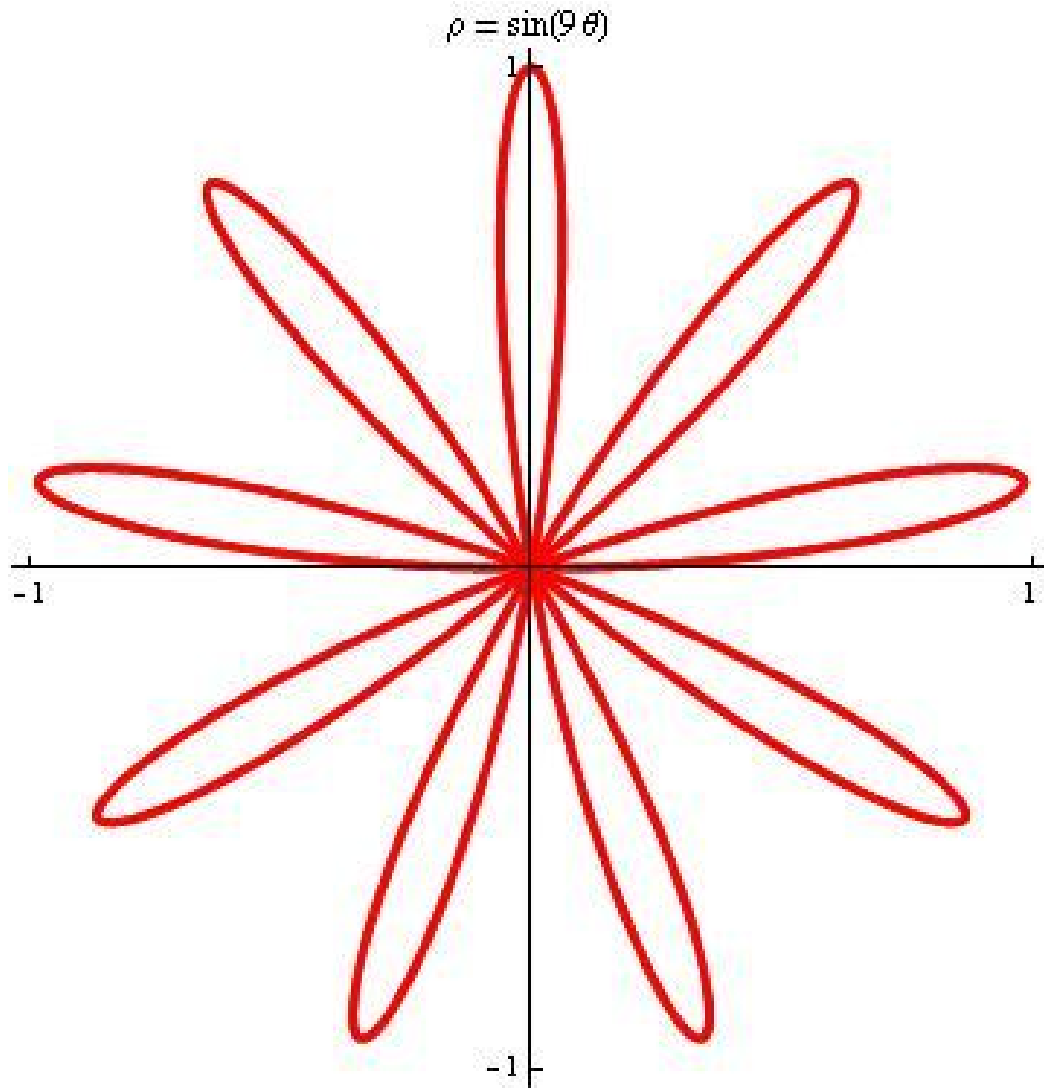
$$\rho = a \cos 4\theta$$



九叶玫瑰线

Nine-leaved rose curve

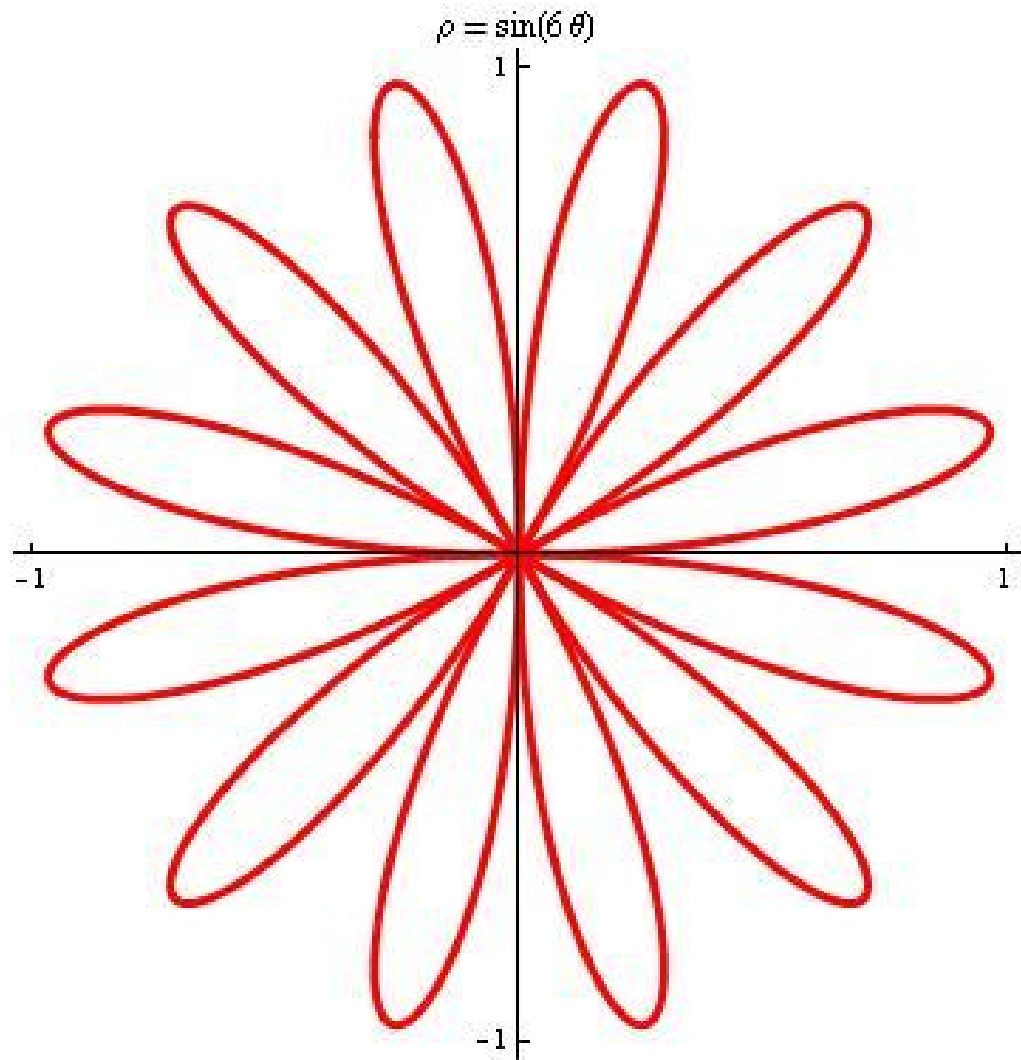
$$\rho = a \sin 9\theta$$



十二叶玫瑰线

Twelve-leaved rose curve

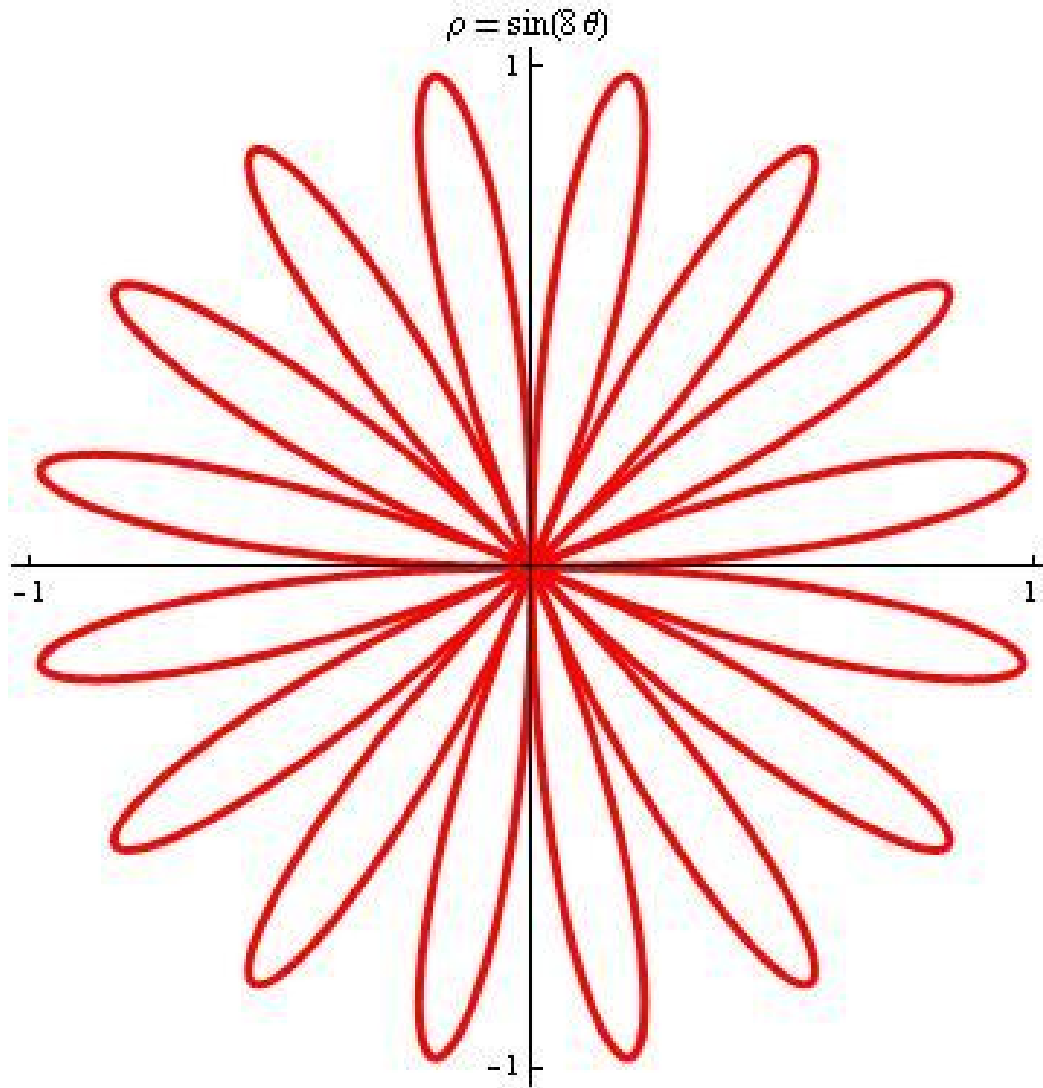
$$\rho = a \sin 6\theta$$



十六叶玫瑰线

Sixteen-leaved rose curve

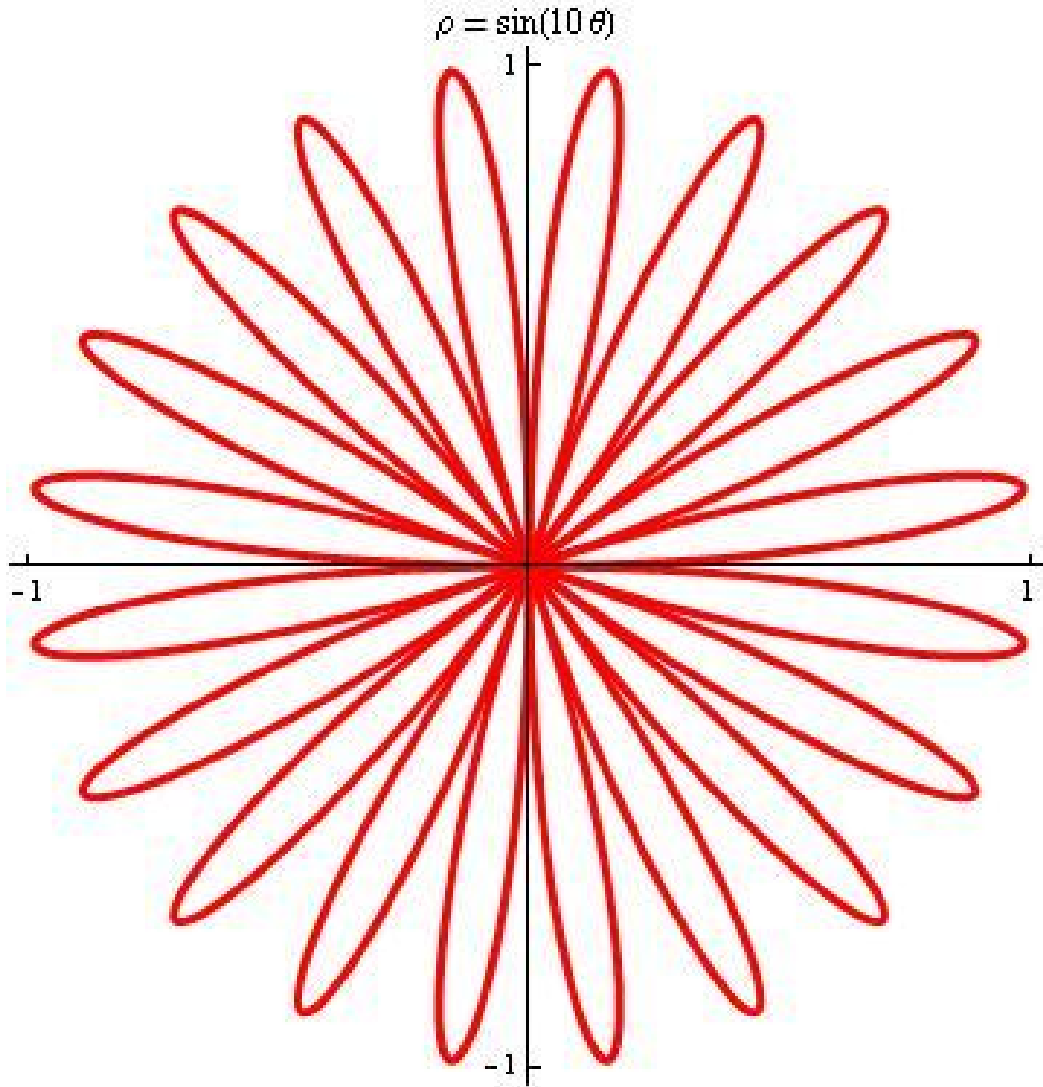
$$\rho = a \sin 8\theta$$



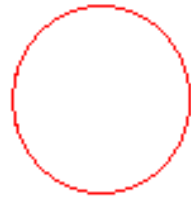
二十叶玫瑰线

Twenty-leaved rose curve

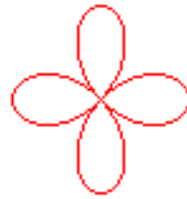
$$\rho = a \sin 10\theta$$



$$\rho = \cos n\theta$$



$n=1$



$n=2$



$n=3$



$n=4$



$n=5$



$n=6$



$n=7$



$n=8$



$n=9$



$n=10$



$n=11$



$n=12$



$n=13$



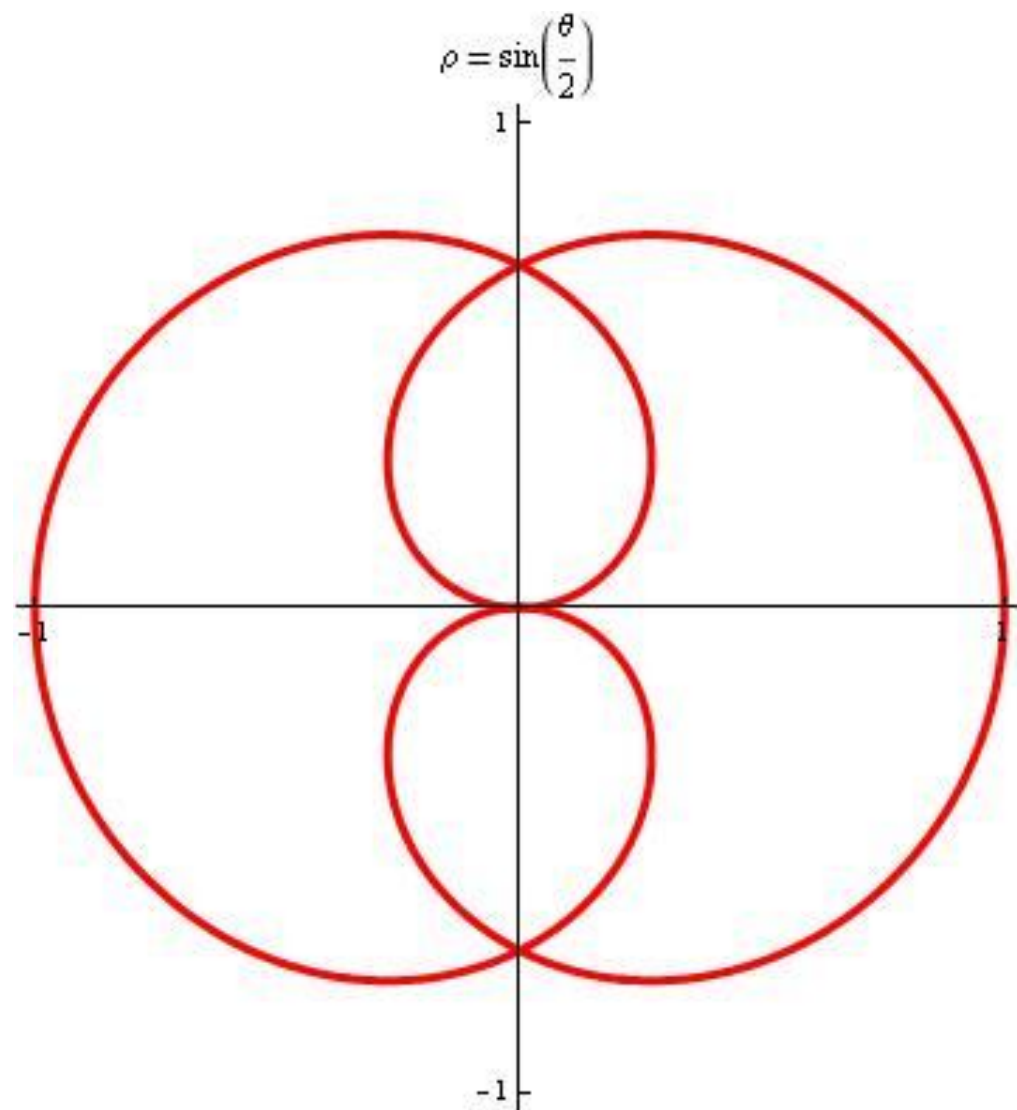
$n=14$



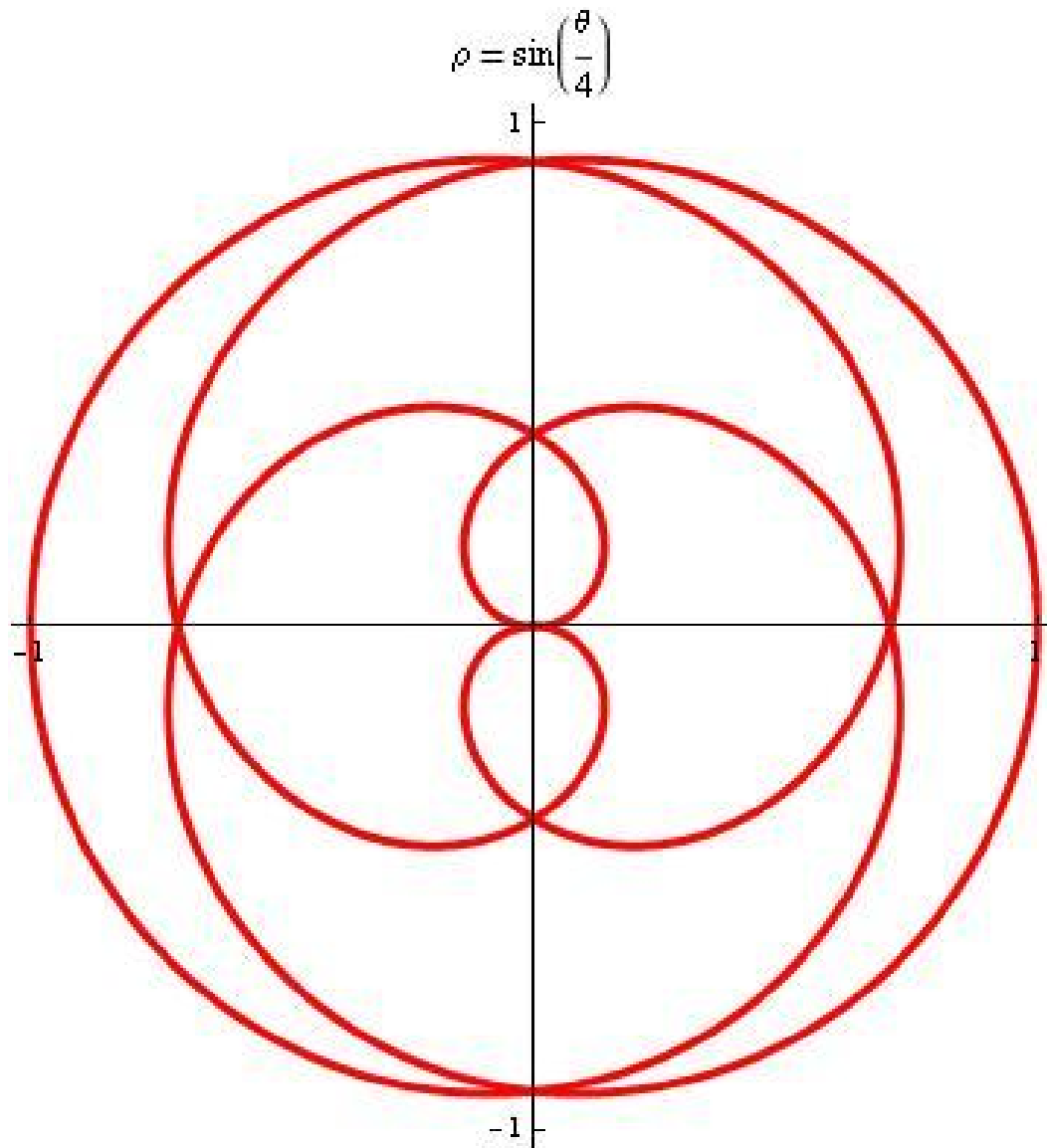
$n=15$



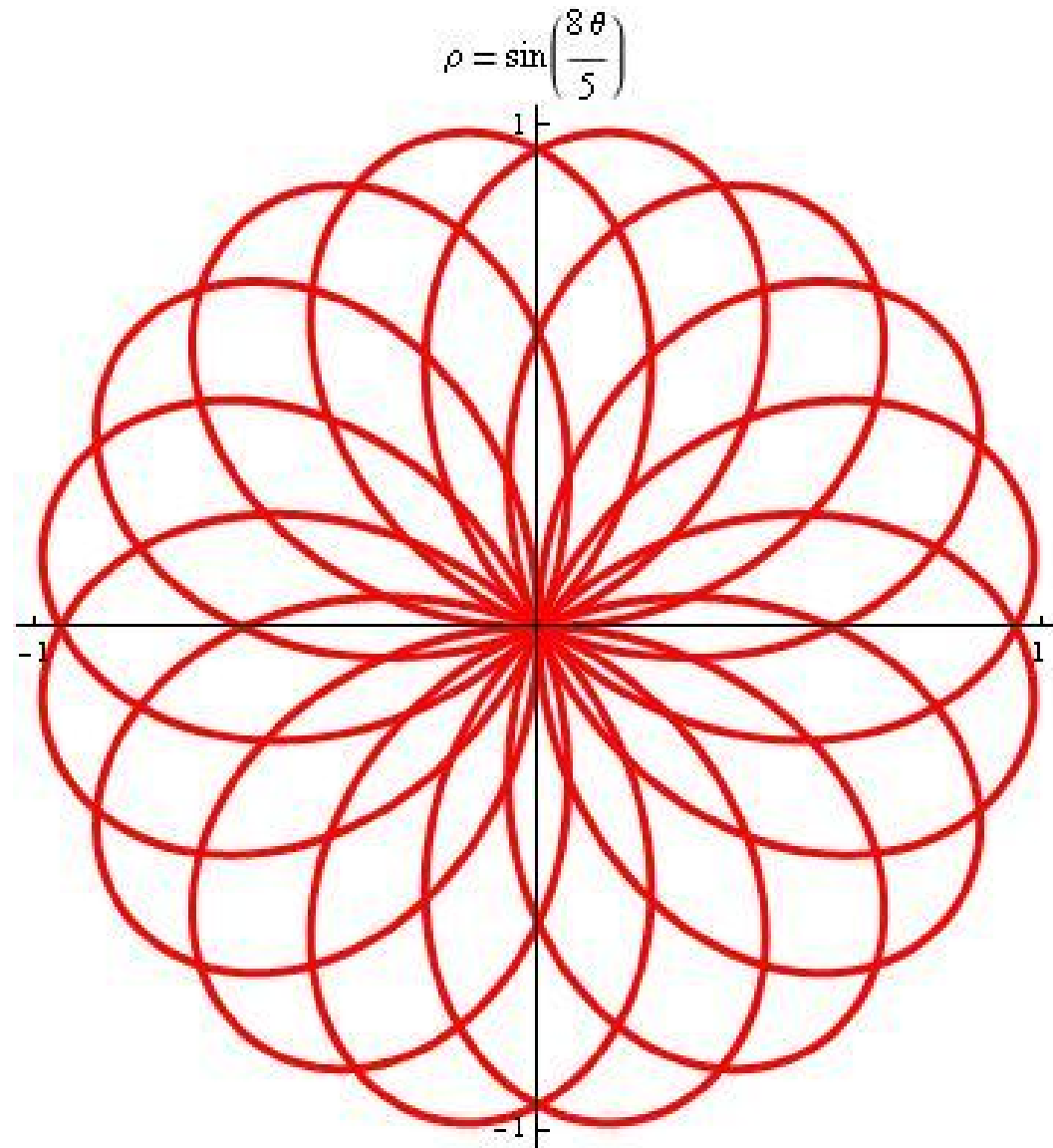
$n=16$



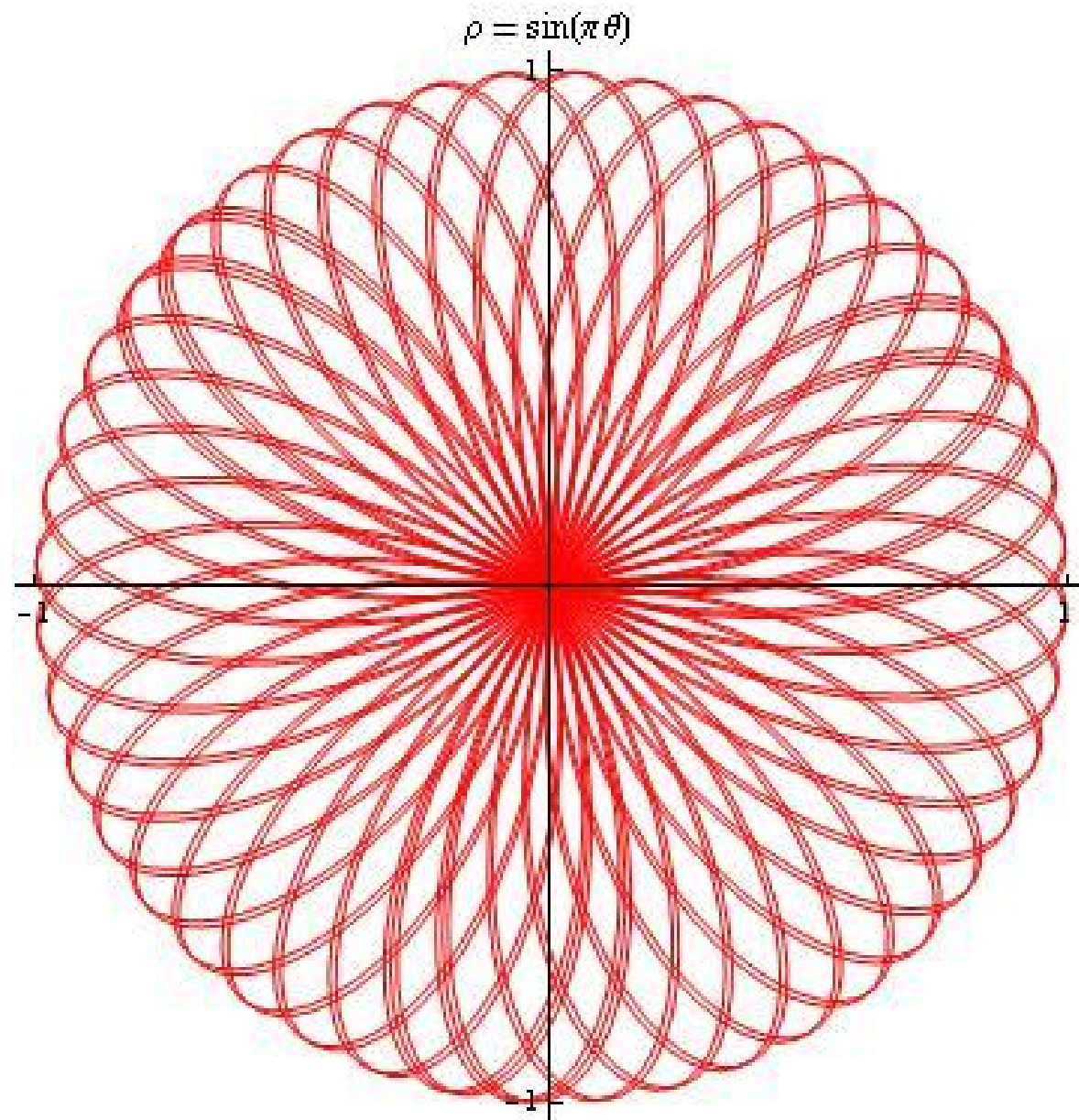
$$\rho = \sin \frac{\theta}{2}$$
$$(0 \leq \theta \leq 4\pi)$$



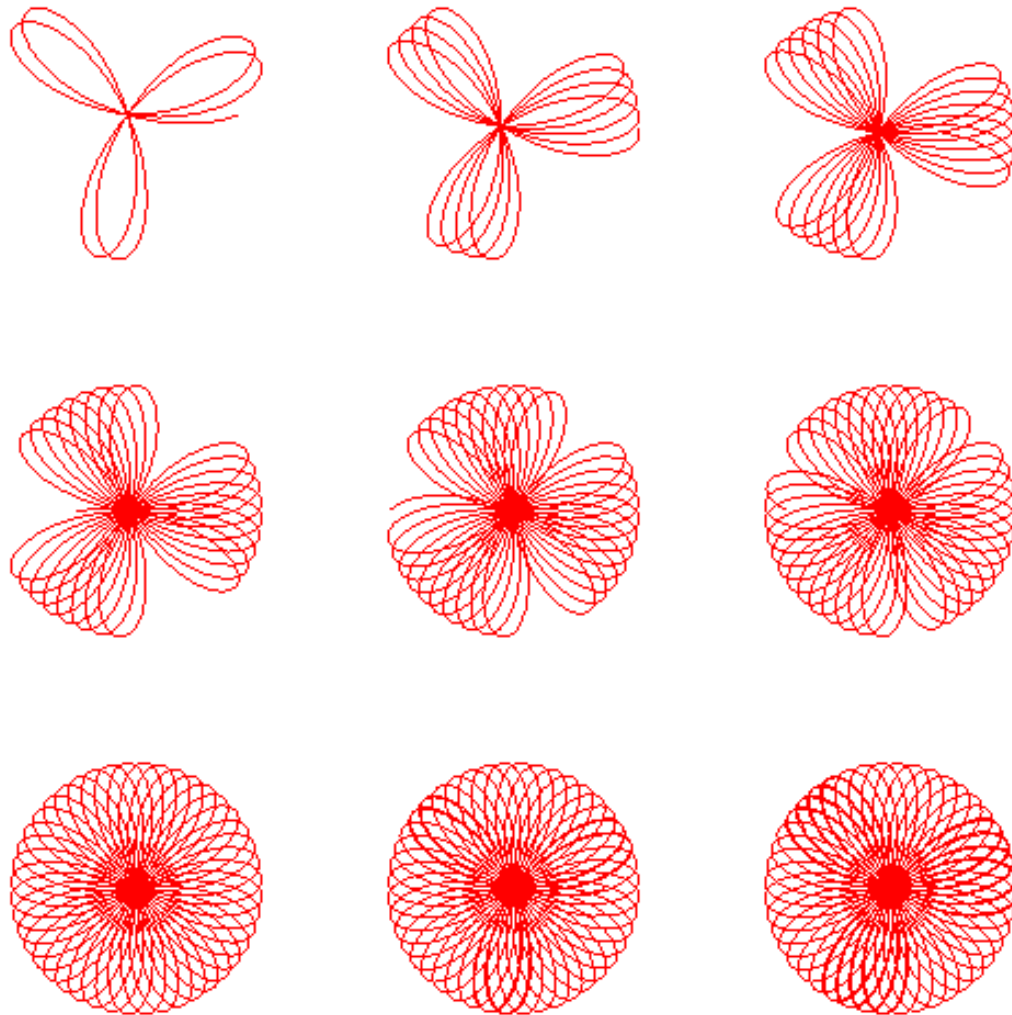
$$\rho = \sin \frac{\theta}{4}$$
$$(0 \leq \theta \leq 8\pi)$$



$$\rho = \sin \frac{8\theta}{5}$$
$$(0 \leq \theta \leq 10\pi)$$



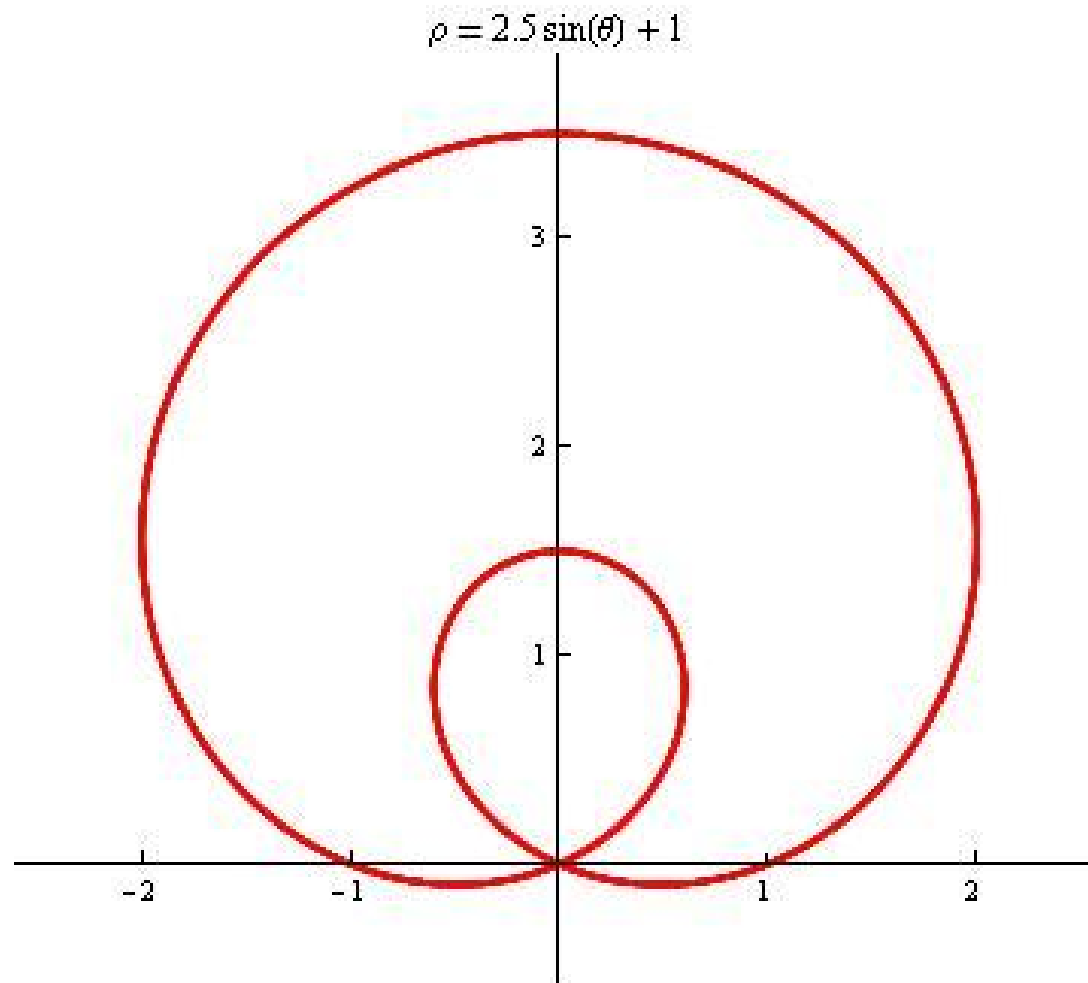
$$\rho = \sin \pi \theta$$



$$\rho = \cos \frac{n}{m} \theta$$

n	m	1	2	3	4	5	6	7
1								
2								
3								
4								
5								
6								
7								

蚶线 (Limacon)



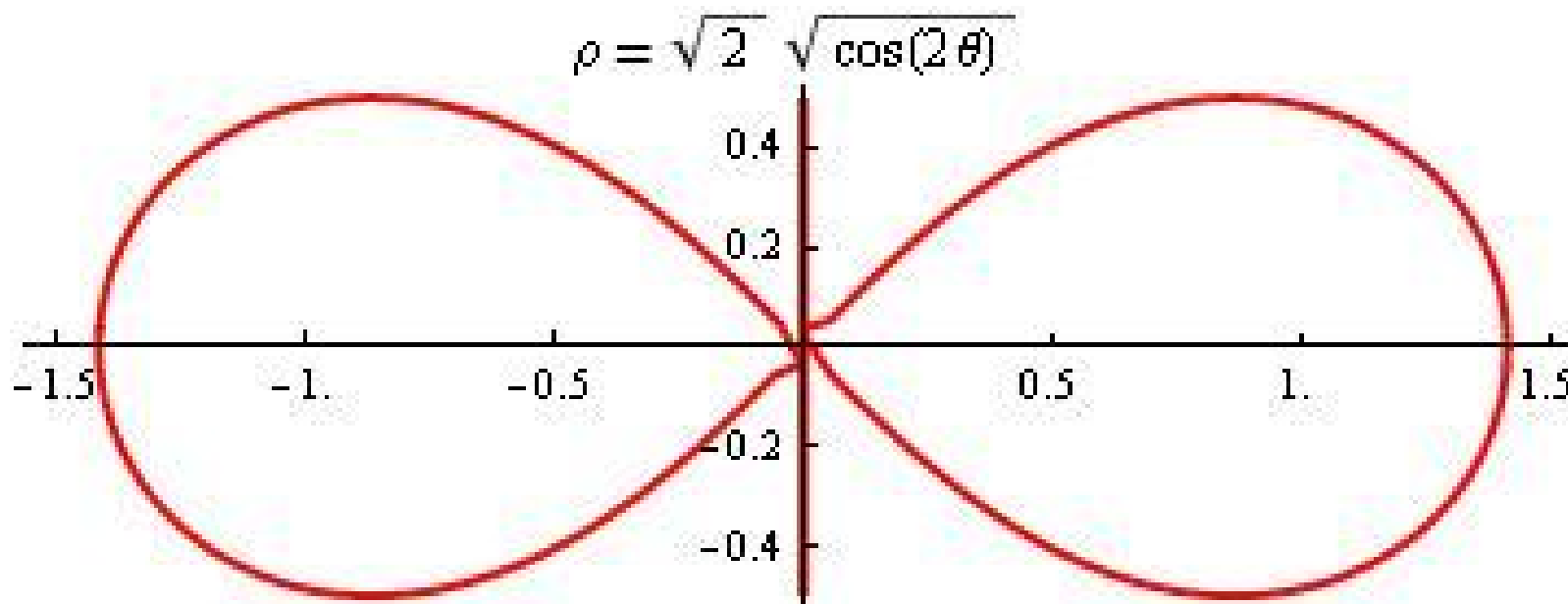
$$\rho = 1 + c \sin \theta$$

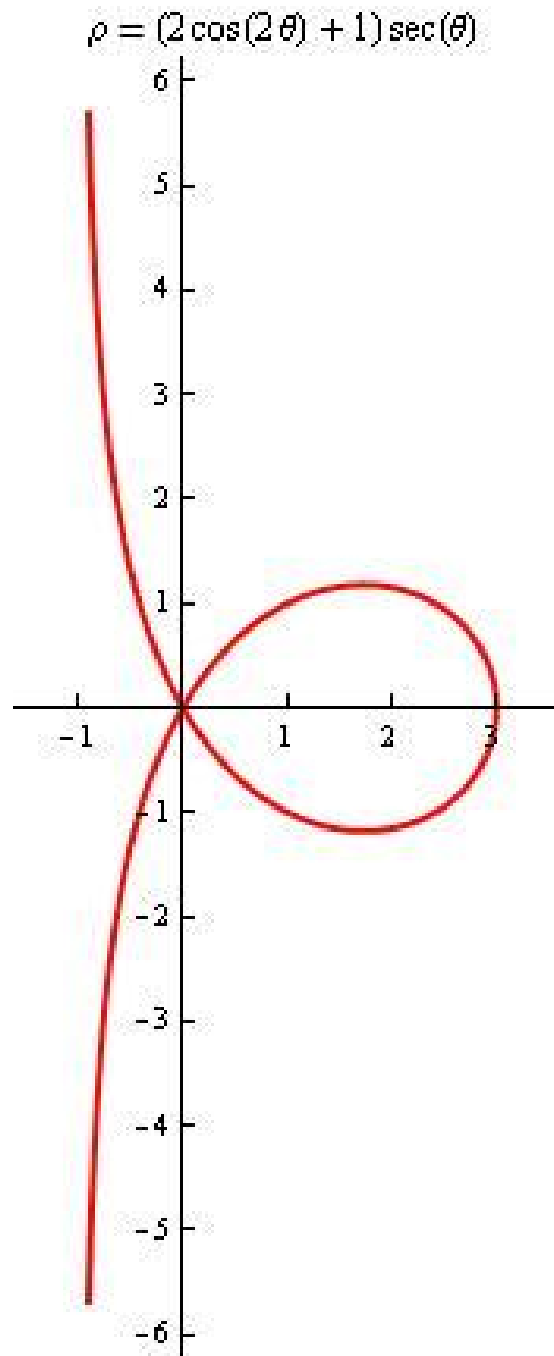
$(c = 2.5)$

双纽线 (lemniscate)

贝努利双纽线

$$\rho^2 = 2a^2 \cos 2\theta$$



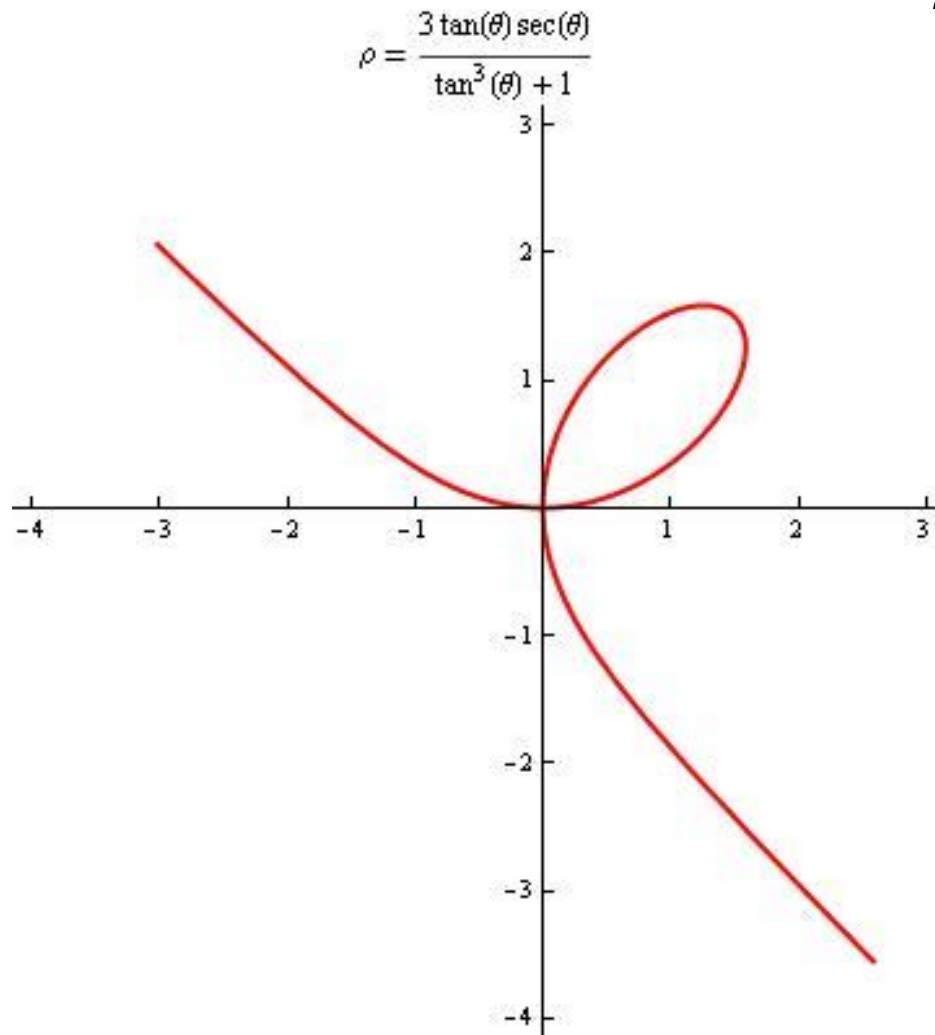


三等分曲线 (**trisectrix**)

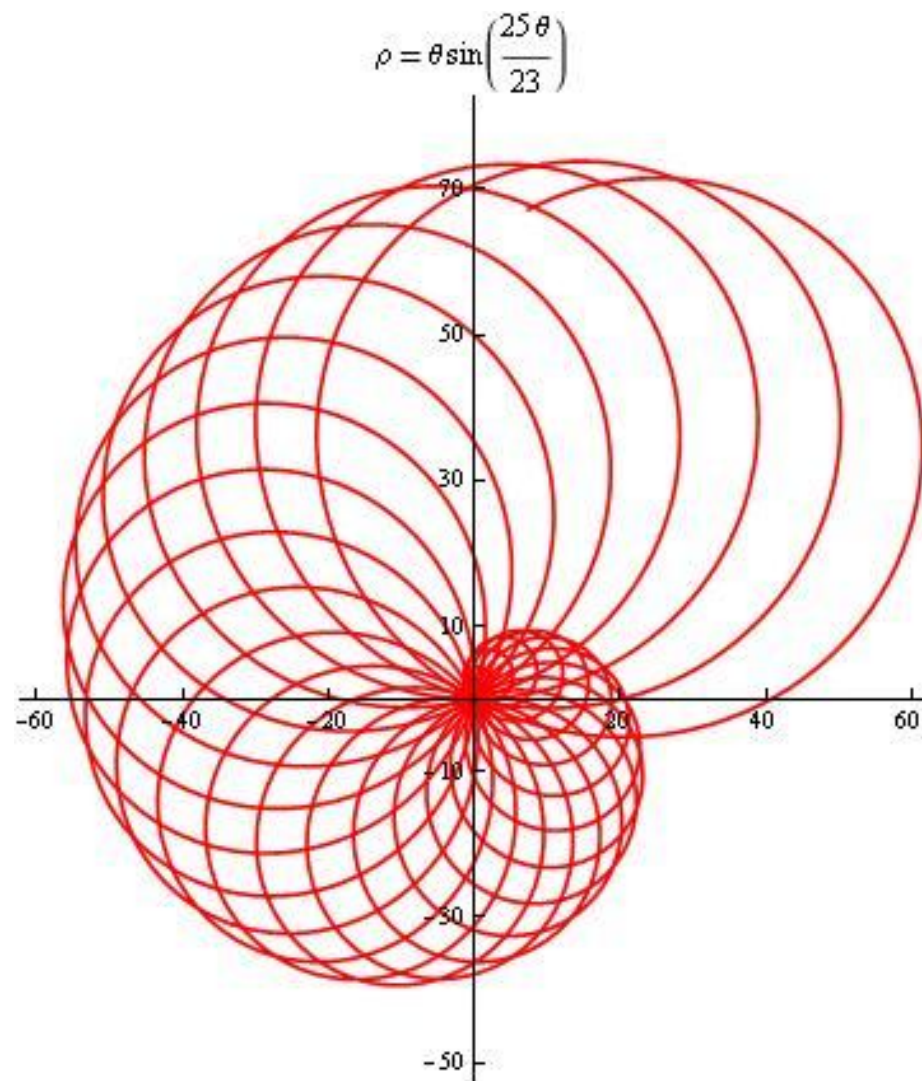
$$\rho = \frac{a(1 + 2 \cos 2\theta)}{\cos \theta}$$

笛卡儿叶形线

(**folium of Descartes**)



$$\rho = \frac{3 \sec \theta \tan \theta}{1 + \tan^3 \theta}$$

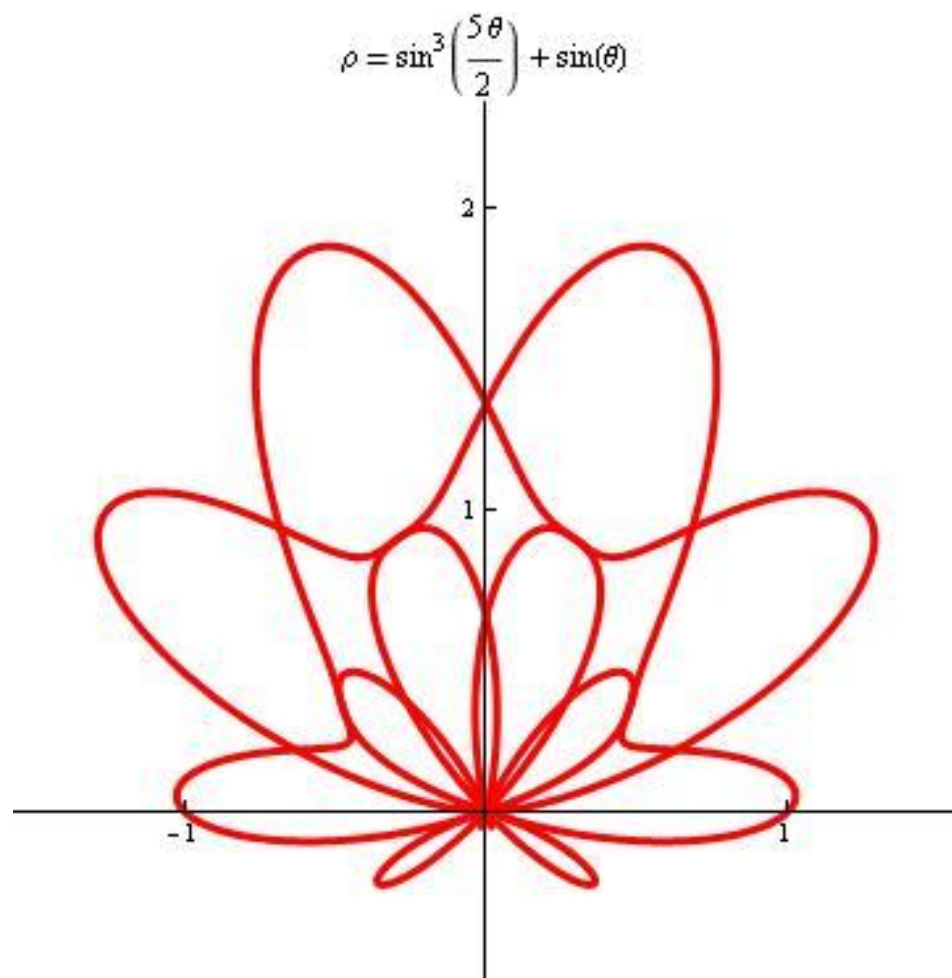


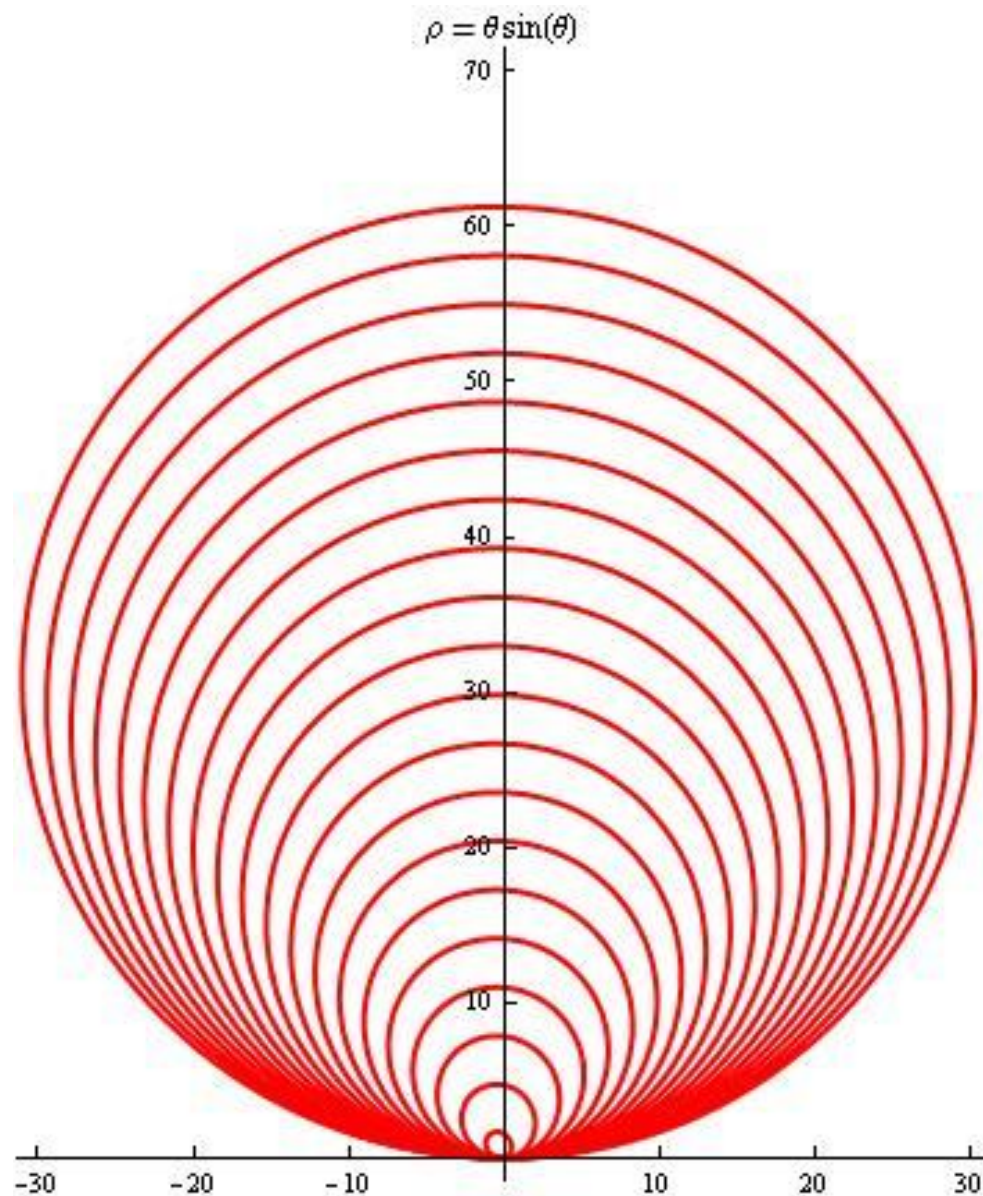
$$\rho = \theta \sin \frac{25\theta}{23}$$

$$(0 \leq \theta \leq 80)$$

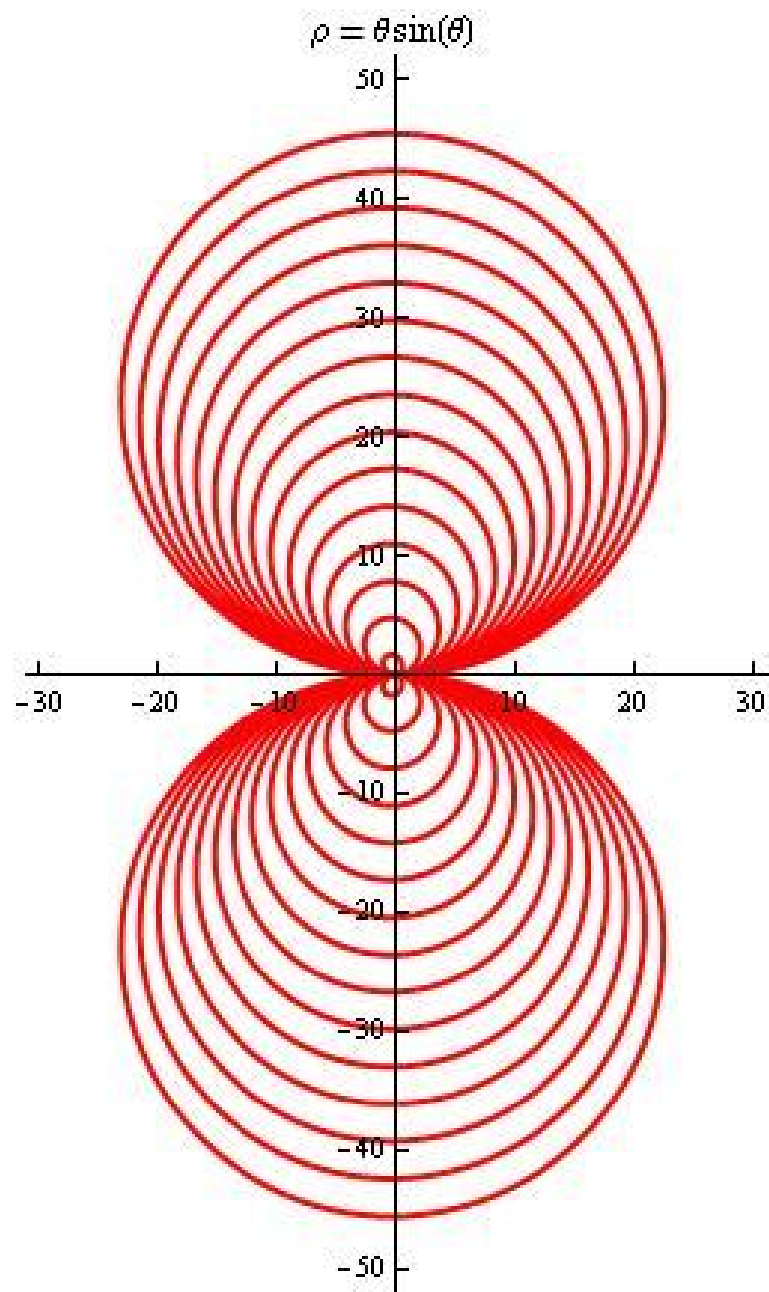
$$\rho = \sin \theta + \sin^3 \frac{5\theta}{2}$$

$$\leq \theta \leq 4\pi)$$





$$\rho = \theta \sin \theta$$
$$(0 \leq \theta \leq 20\pi)$$



$$\rho = \theta \sin \theta$$
$$(-15\pi \leq \theta \leq 15\pi)$$